



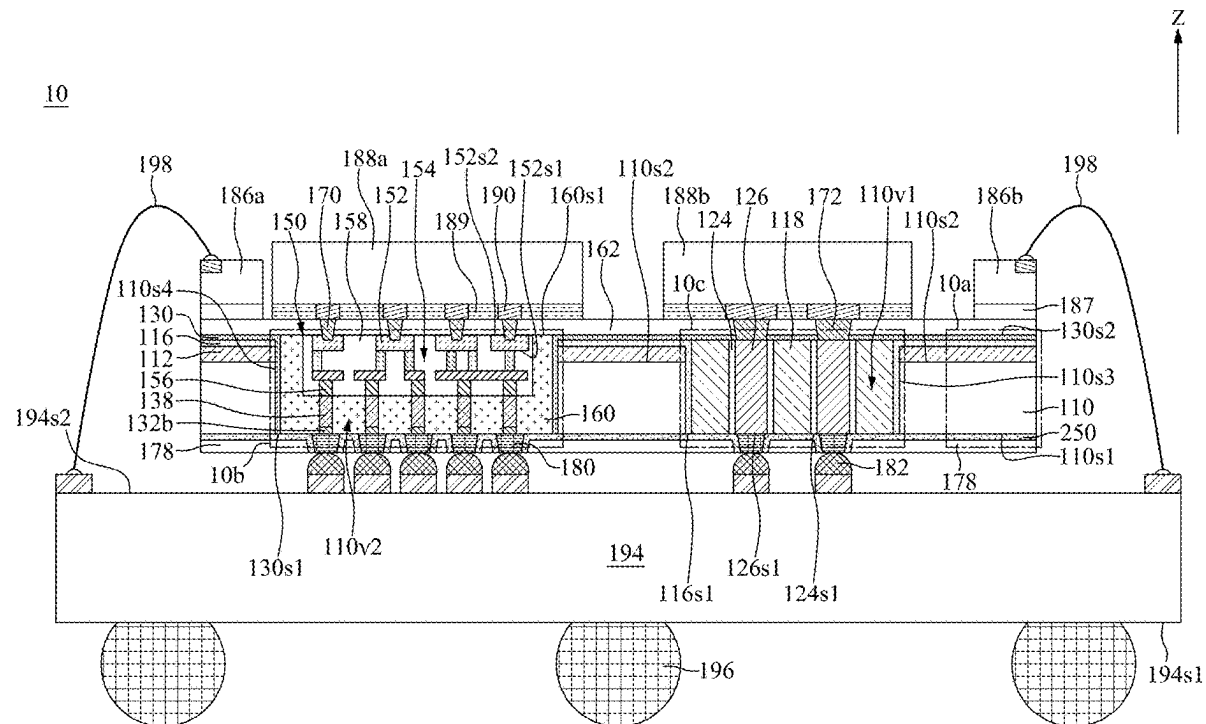
US 20250286026A1

(19) **United States**(12) **Patent Application Publication**
SU(10) **Pub. No.: US 2025/0286026 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SEMICONDUCTOR DEVICE WITH SPACER
AND METHOD FOR FABRICATING THE
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H01L 23/498 (2006.01)
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H01L 24/48 (2013.01); **H01L 2224/08145**
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(57)

ABSTRACT

The present application discloses a semiconductor device and a method for fabricating the semiconductor device. The semiconductor device includes an interposer having a first surface and a second surface parallel to the first surface; a conductive via extending between the first surface and the second surface of the interposer; an insulation layer separating the conductive via from the interposer; a first electronic component positioned on the second surface and electrically connected to the conductive via; a first conductive element positioned on the first surface of the interposer and electrically connected to the conductive via; a spacer conformally positioned on the first surface of the interposer and on sidewalls of the first conductive element; and a passivation layer positioned on the spacer. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.



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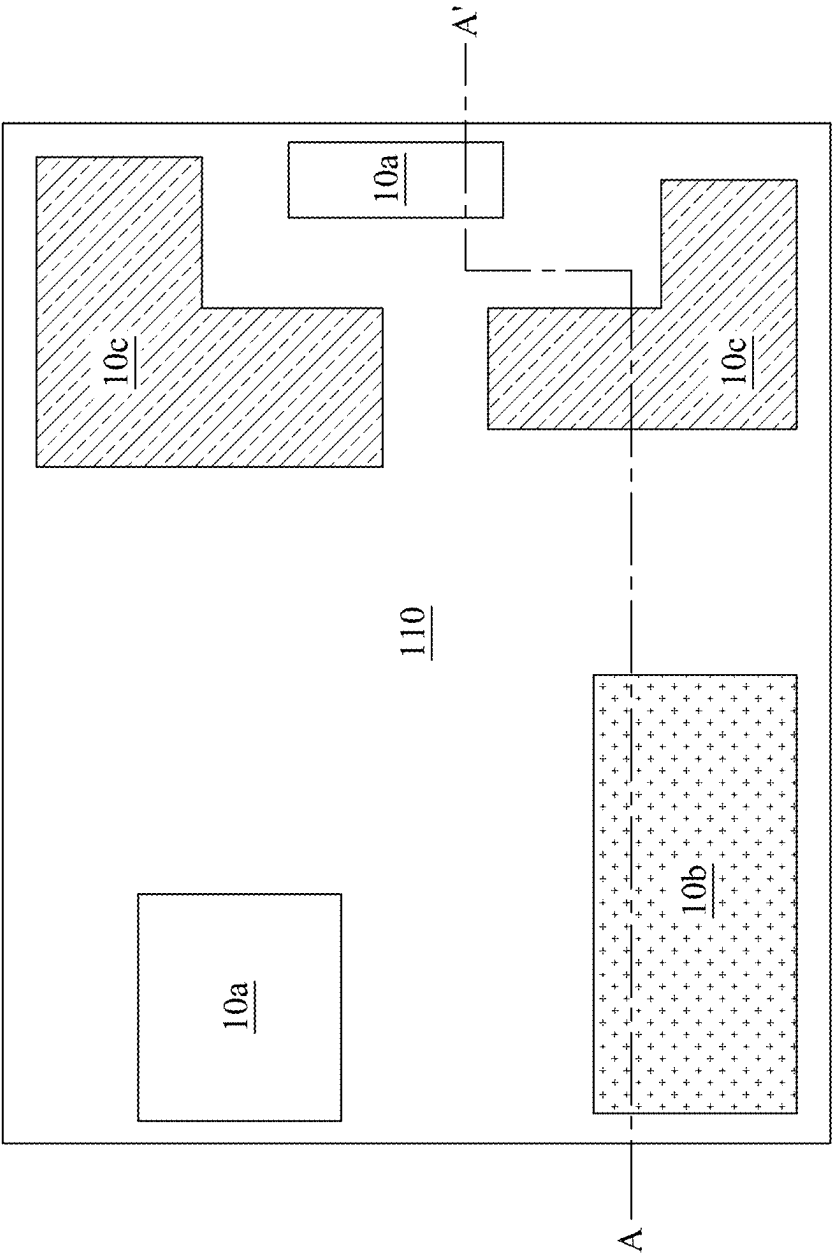


FIG. 1

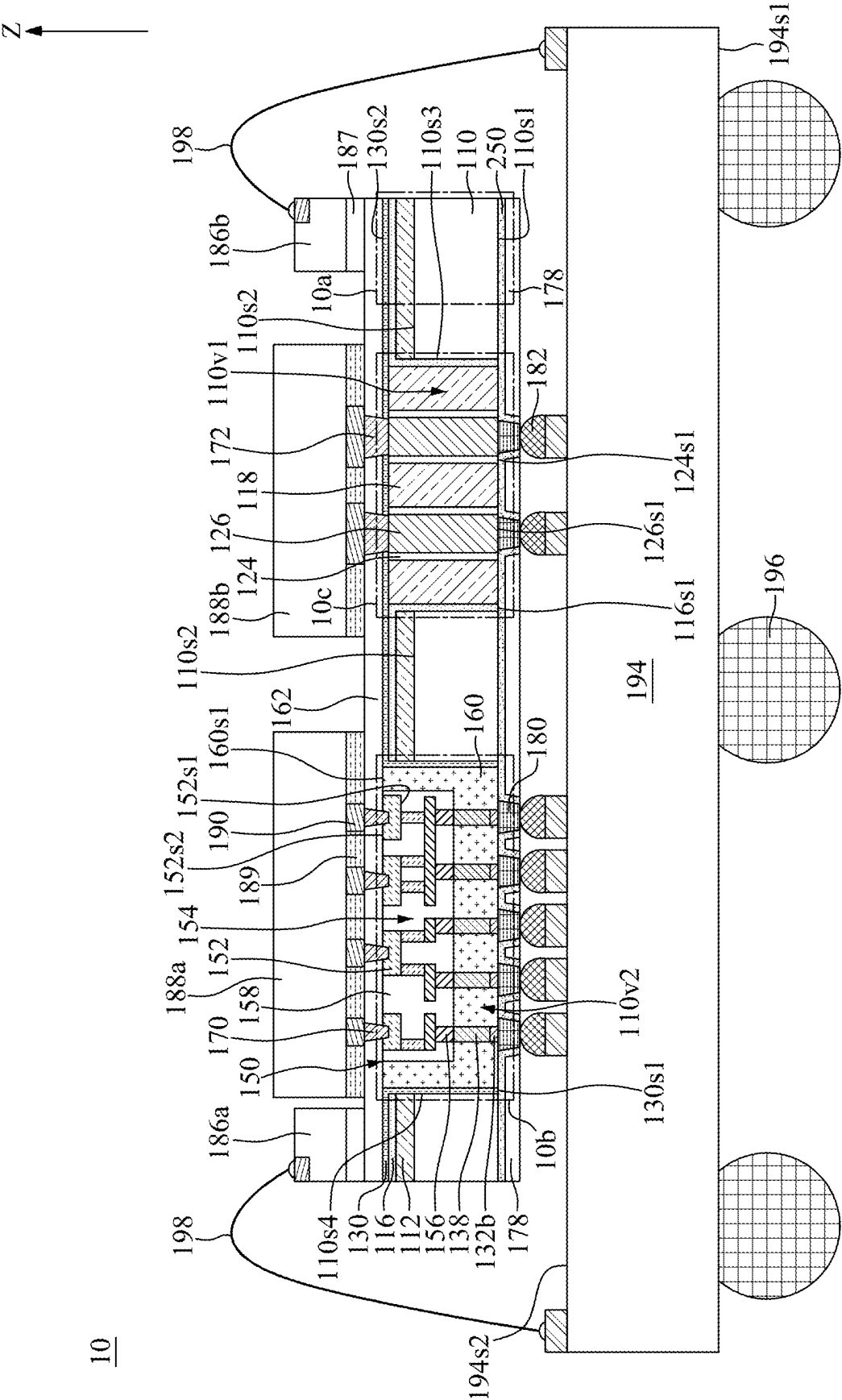


FIG. 2

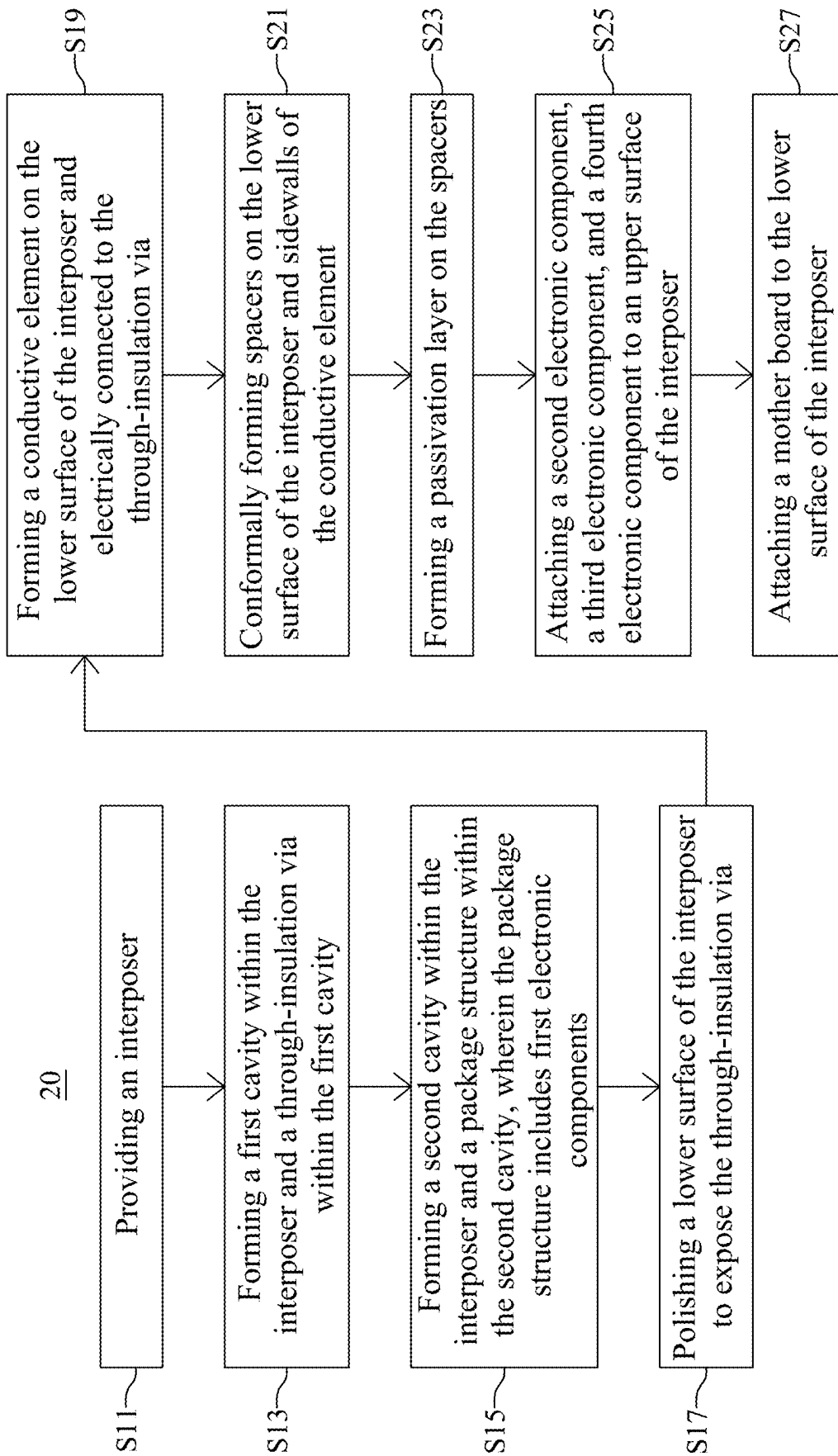


FIG. 5

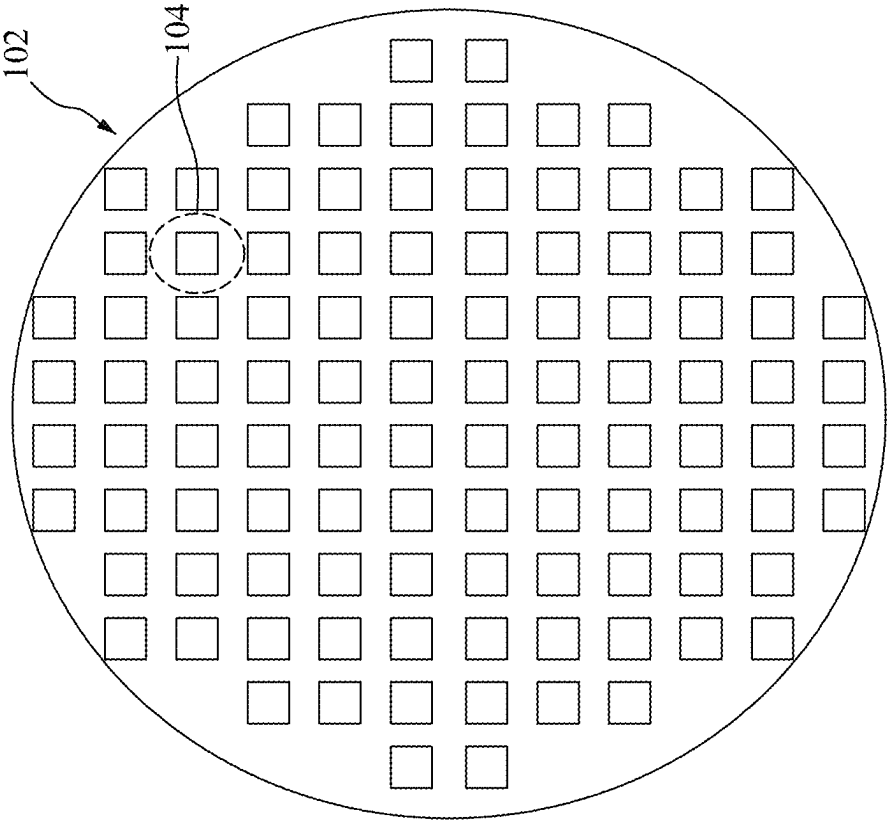


FIG. 6

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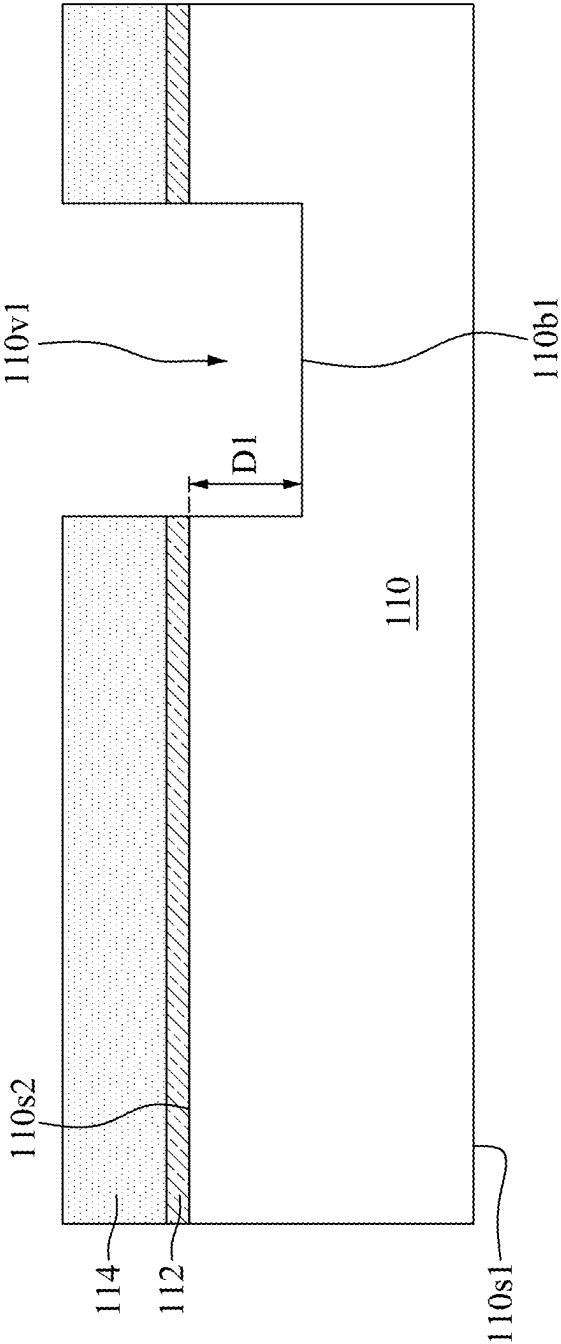


FIG. 7

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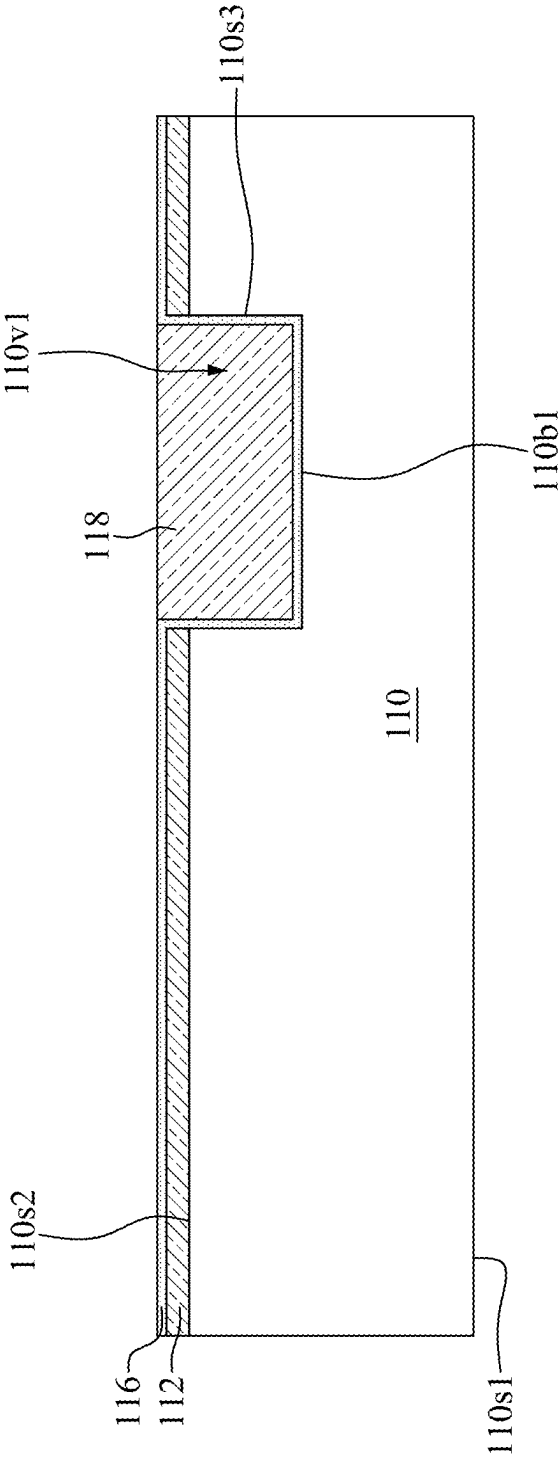


FIG. 8

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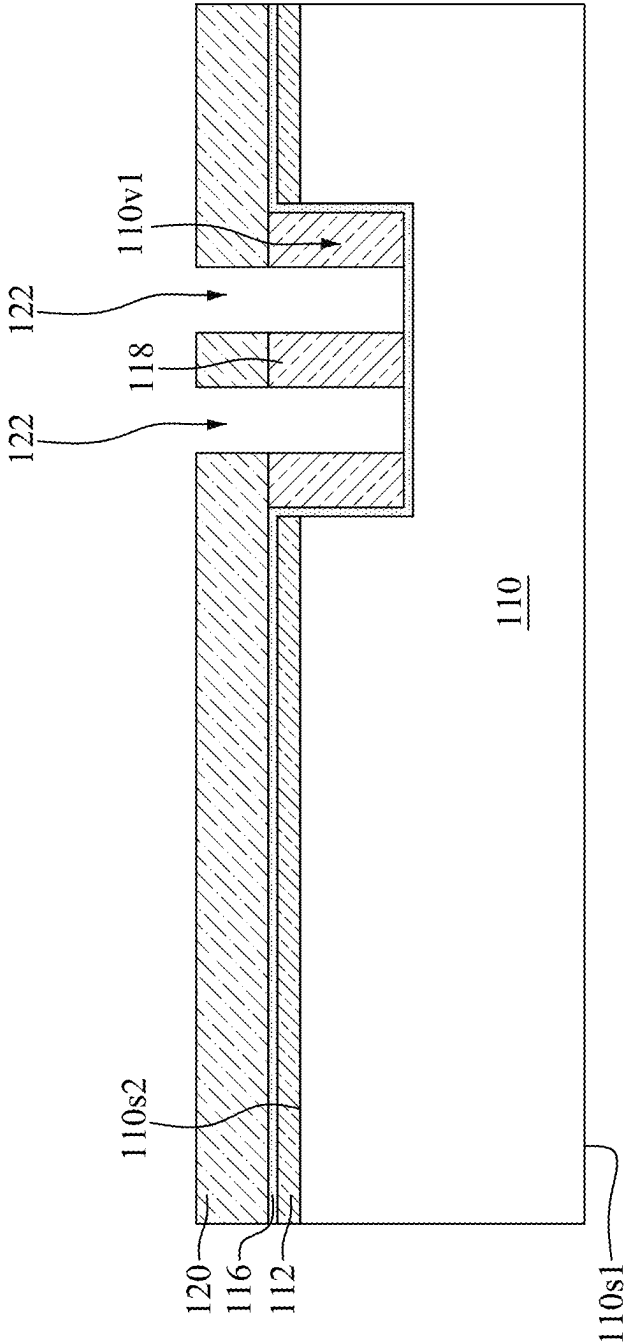


FIG. 9

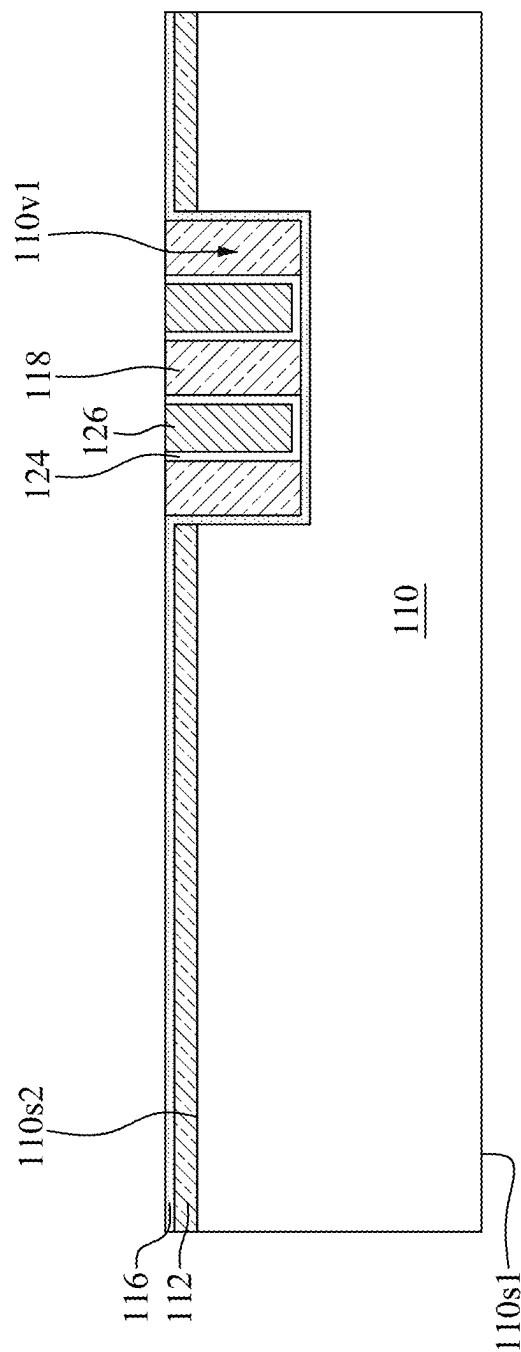
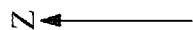


FIG. 10

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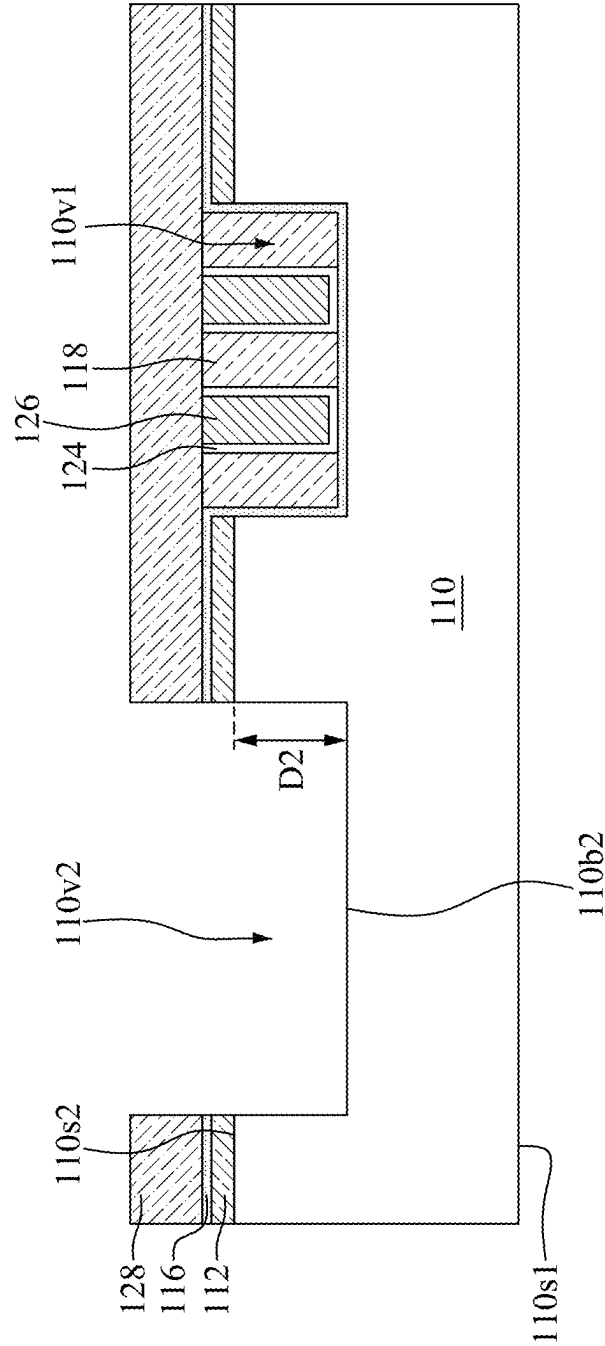


FIG. 11

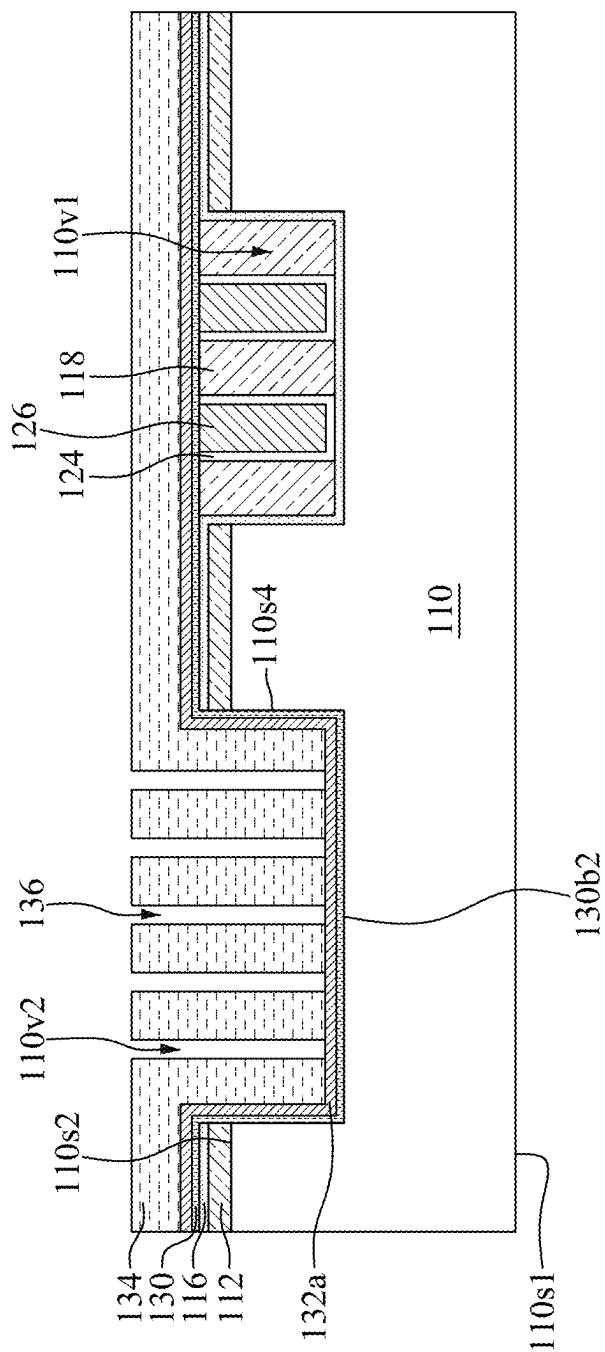
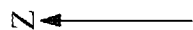


FIG. 12

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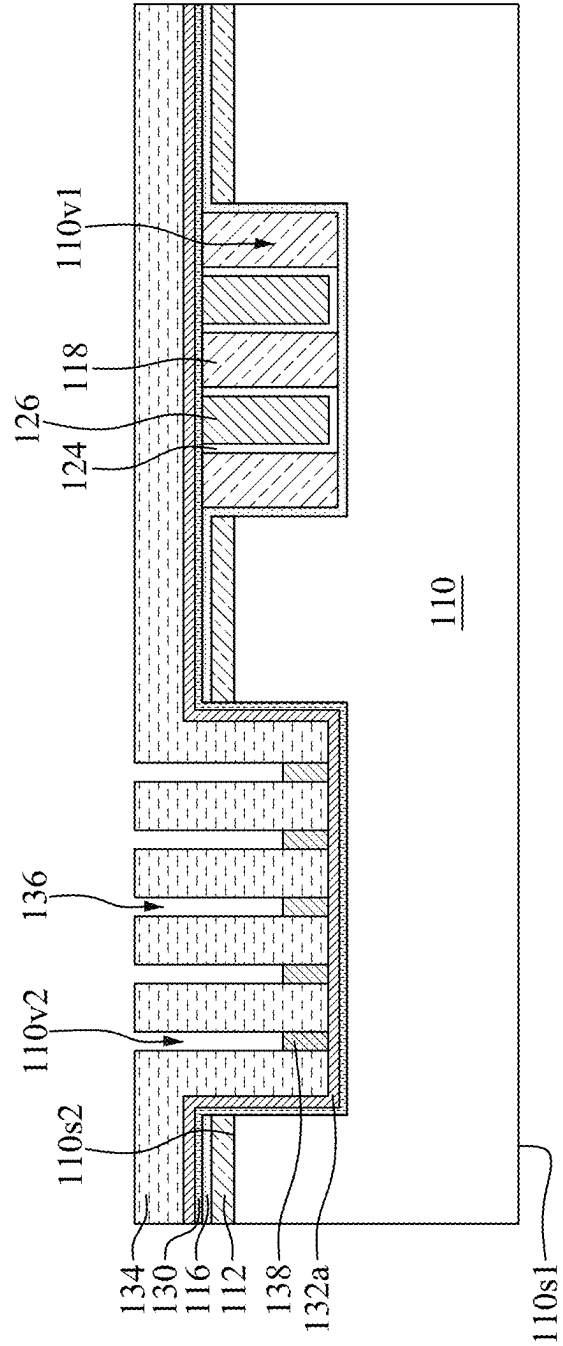


FIG. 13

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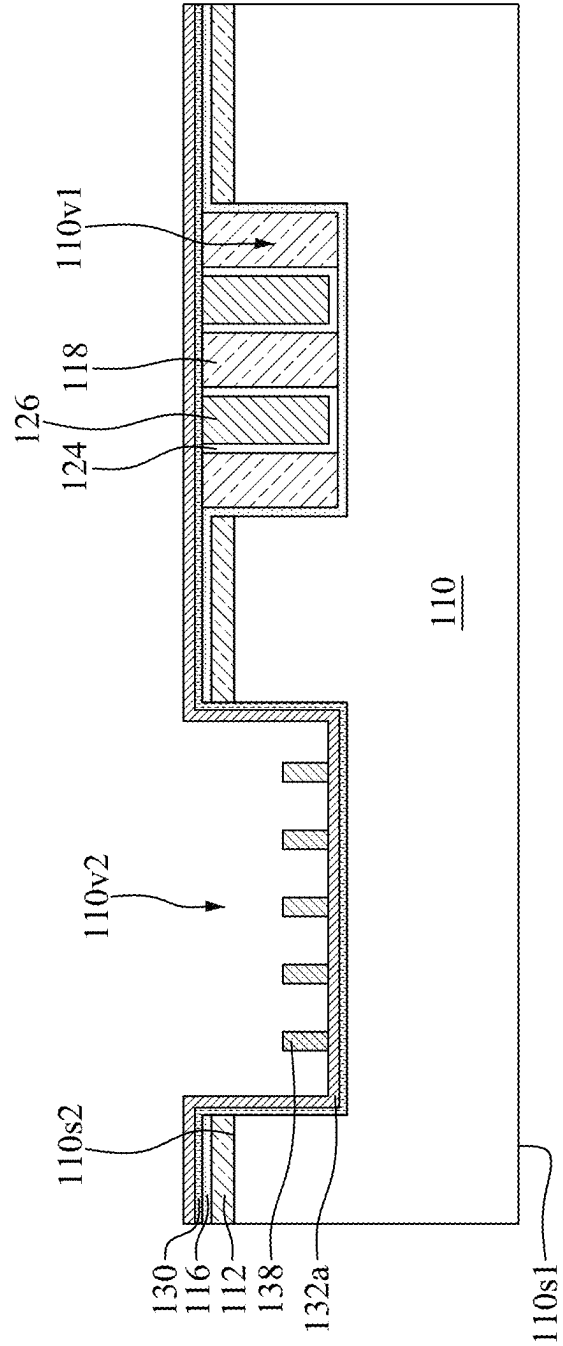


FIG. 14

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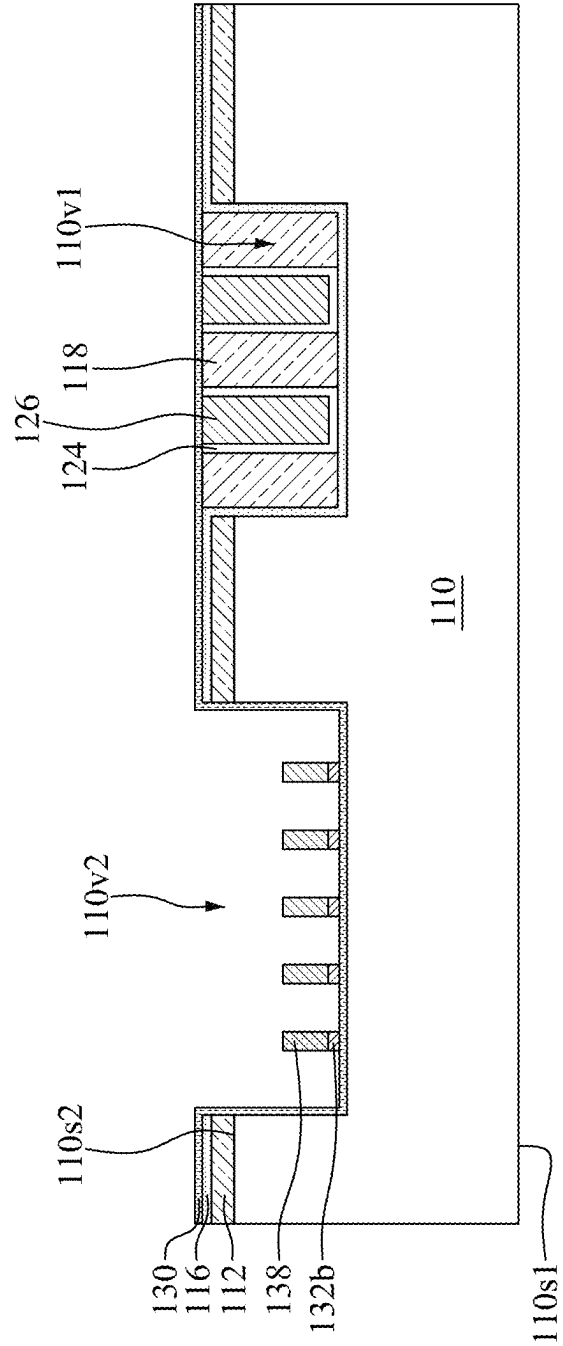


FIG. 15

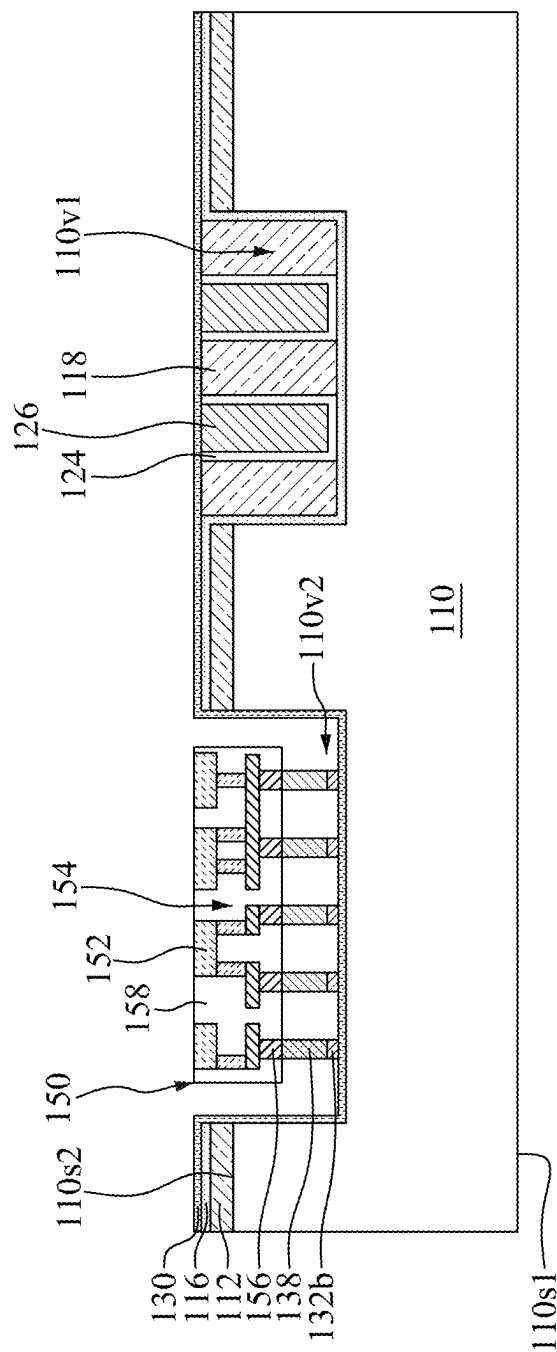
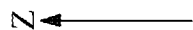


FIG. 16

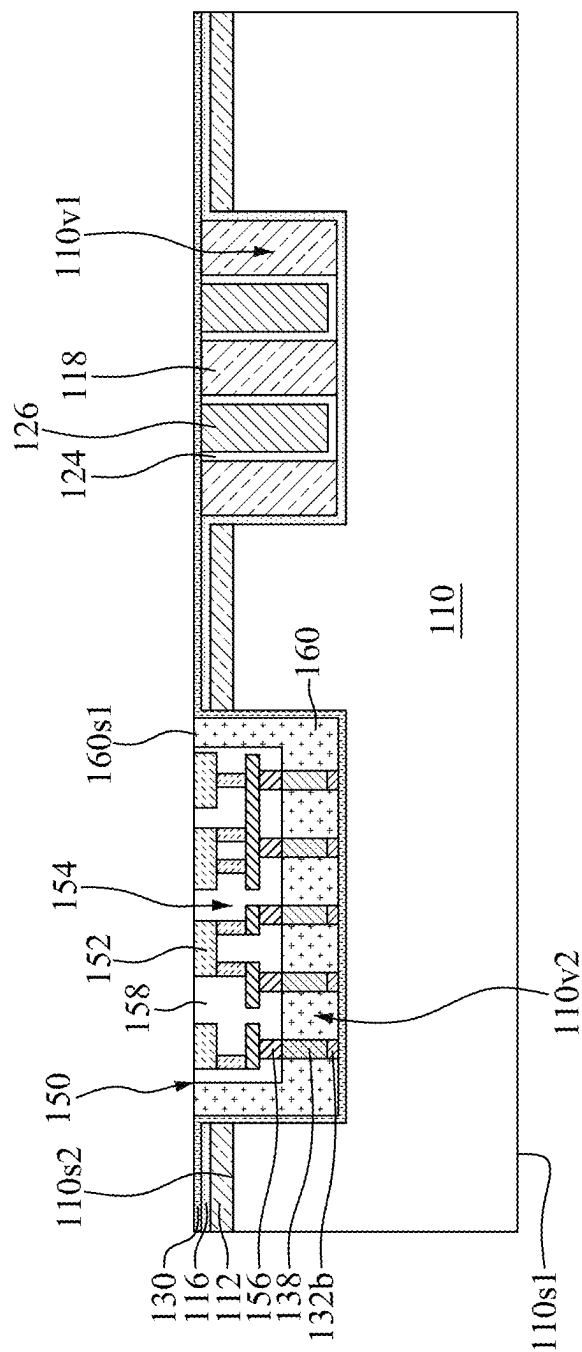
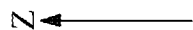


FIG. 17

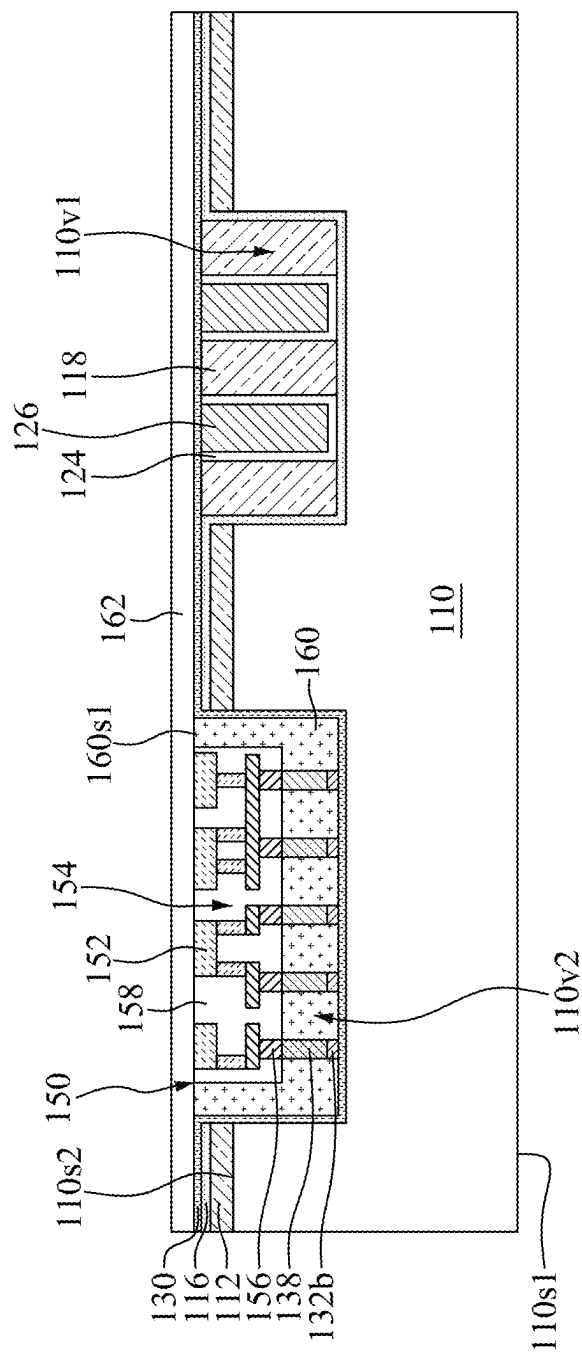
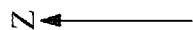


FIG. 18

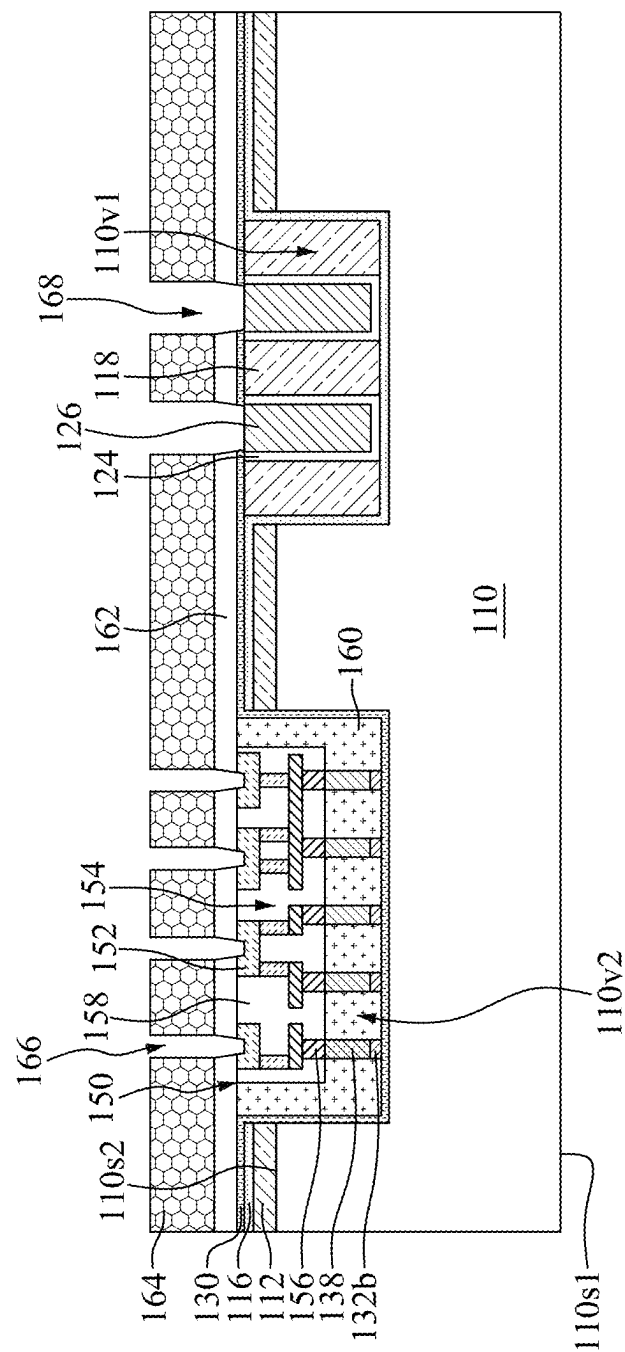
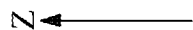


FIG. 19

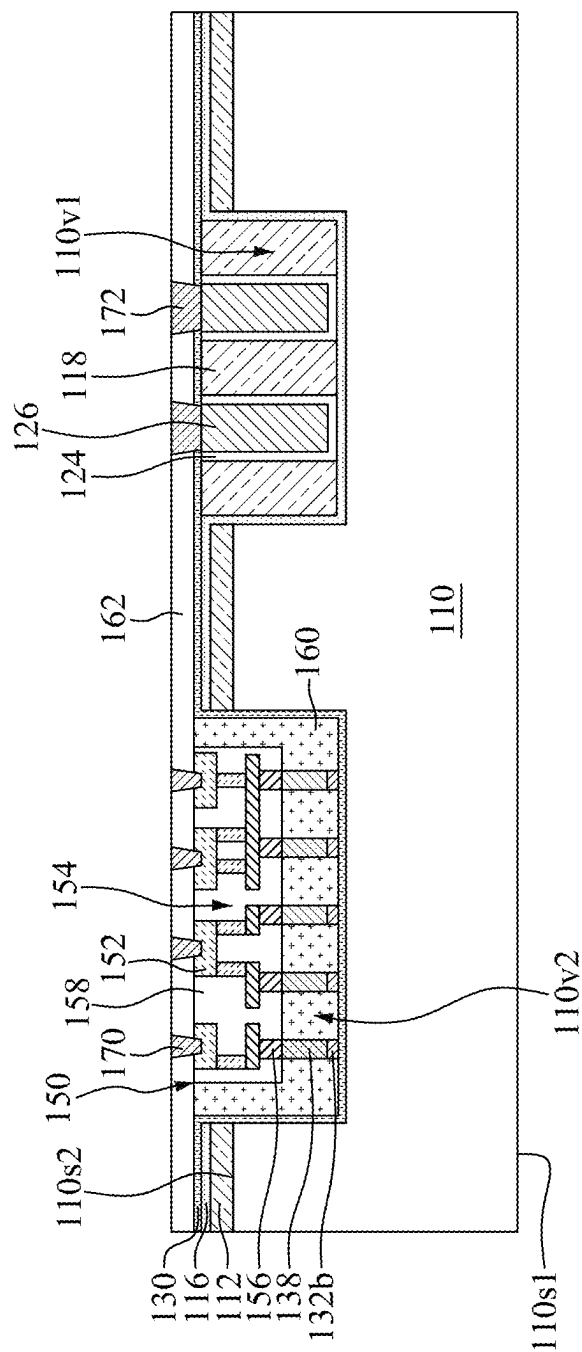
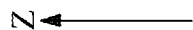


FIG. 20

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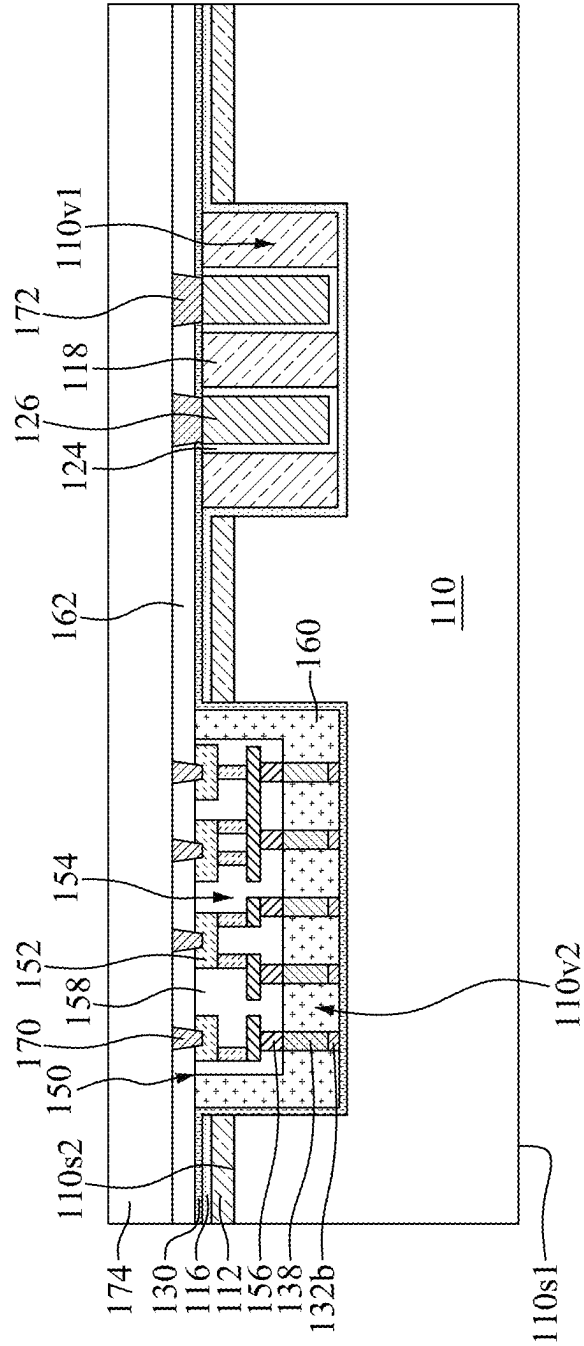


FIG. 21

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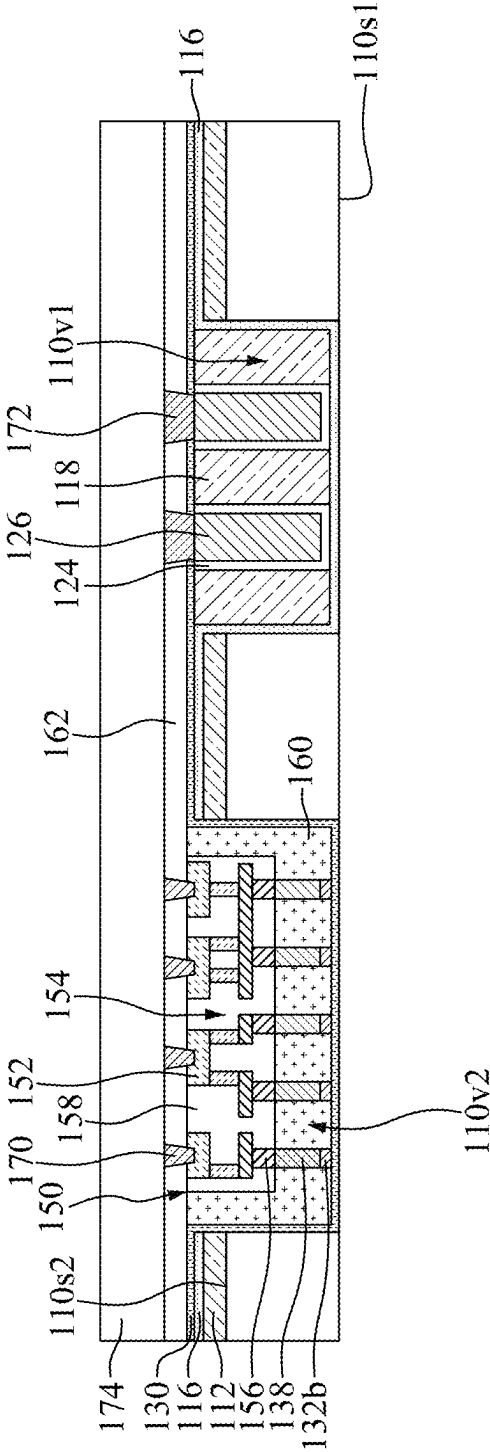


FIG. 22

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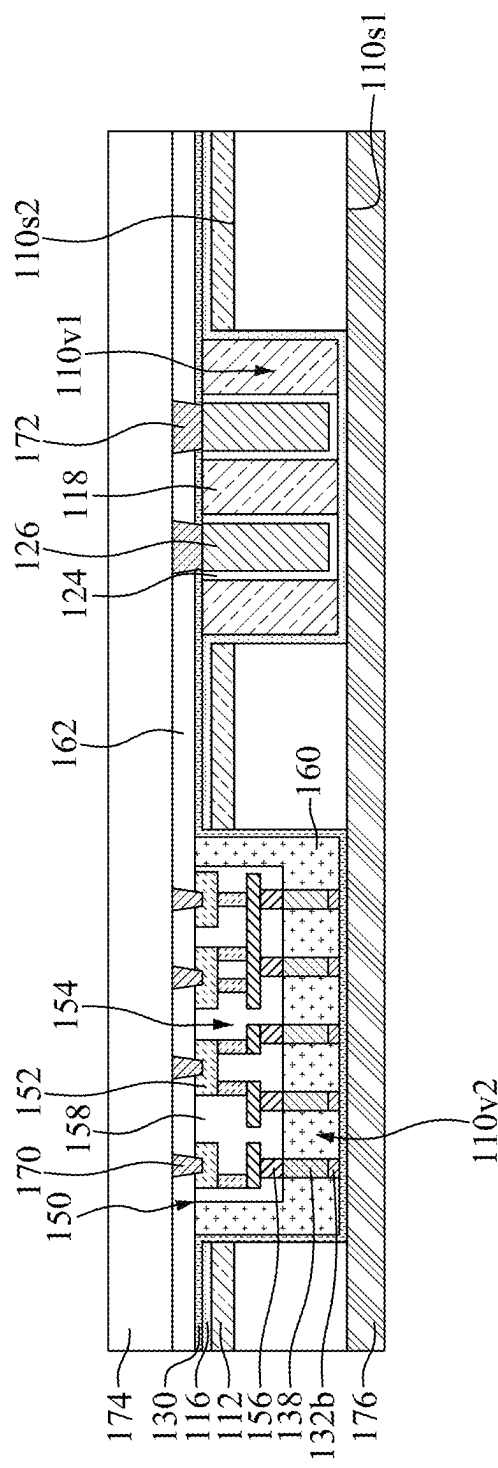


FIG. 23

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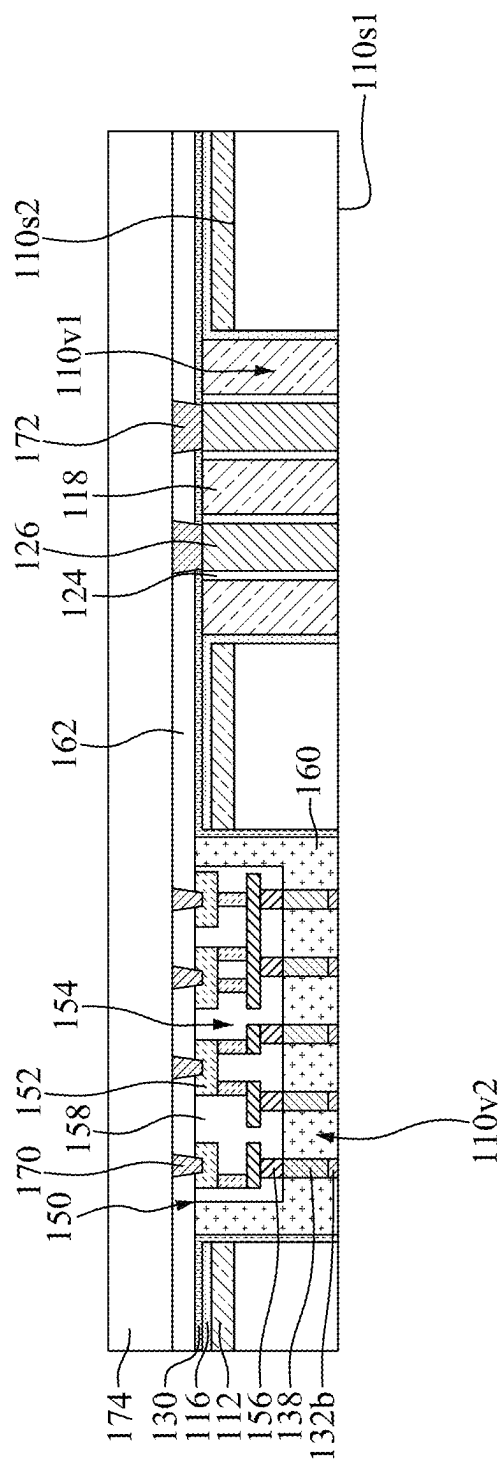


FIG. 24

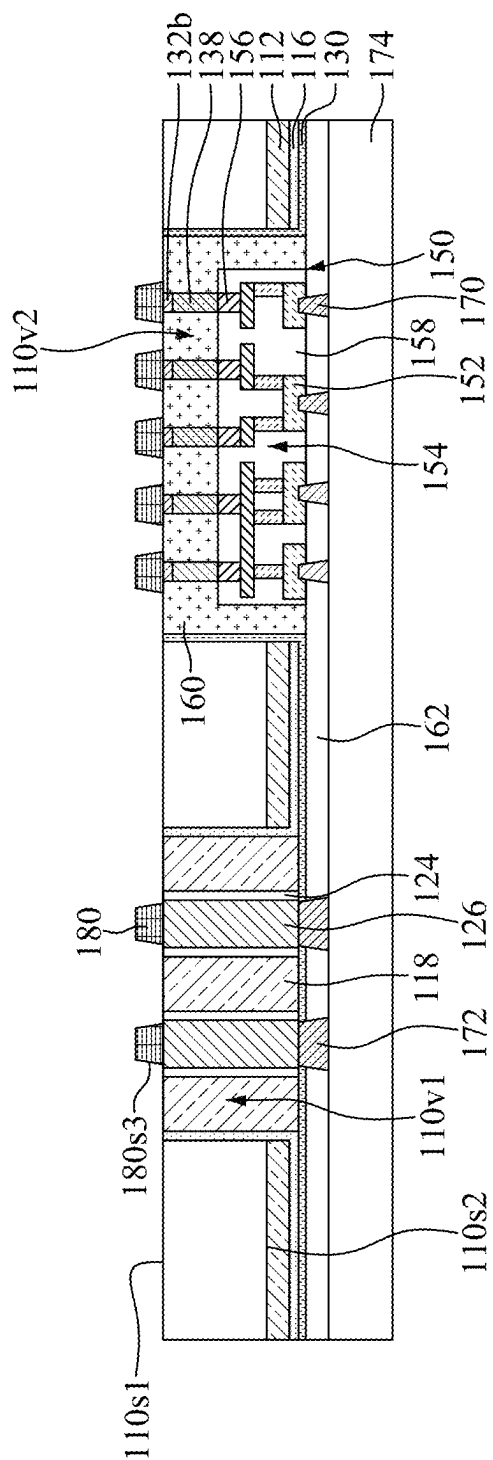
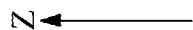


FIG. 25

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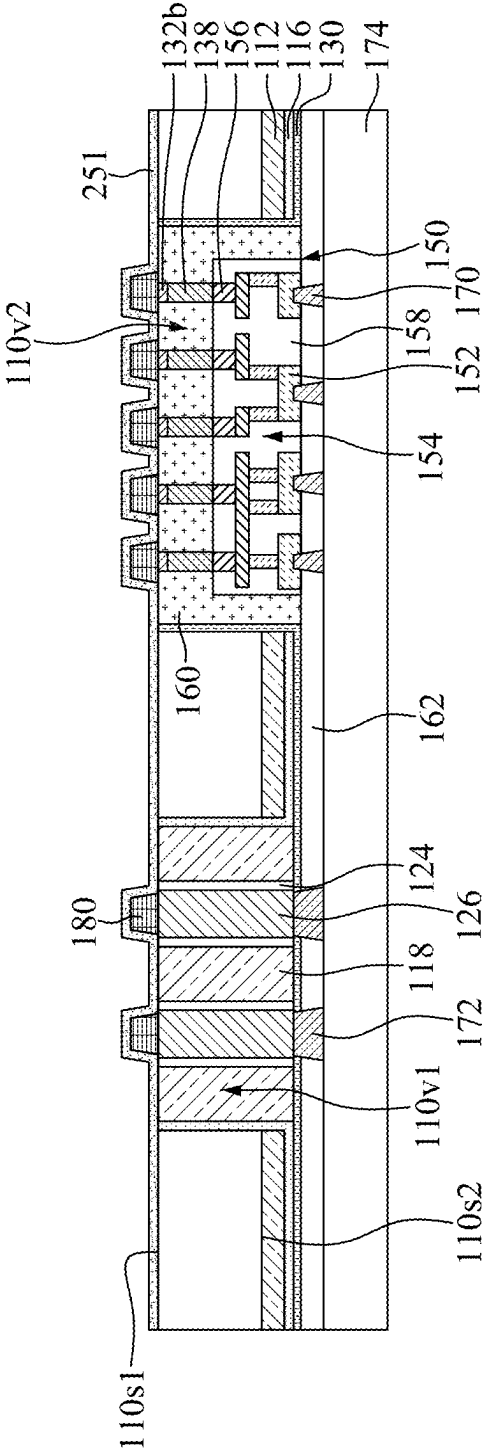


FIG. 26

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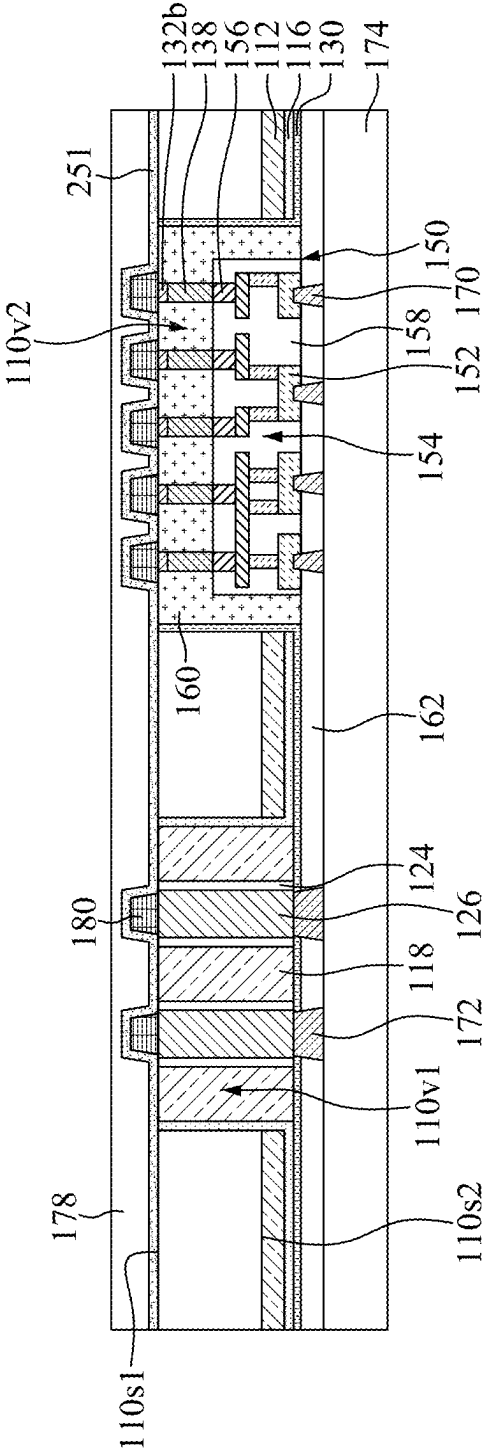


FIG. 27

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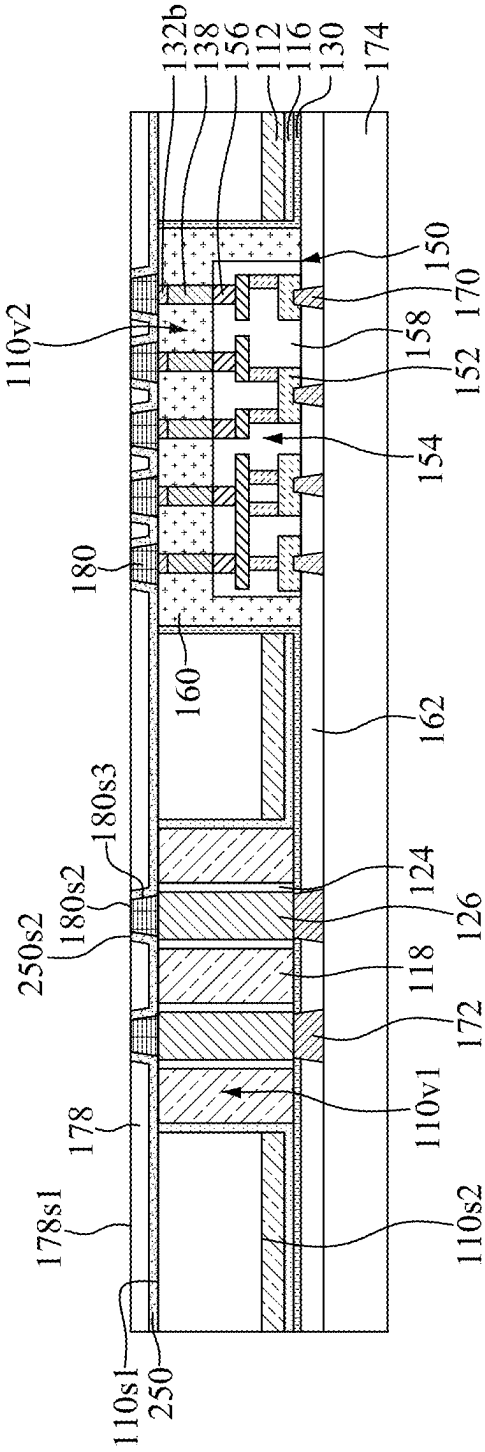


FIG. 28

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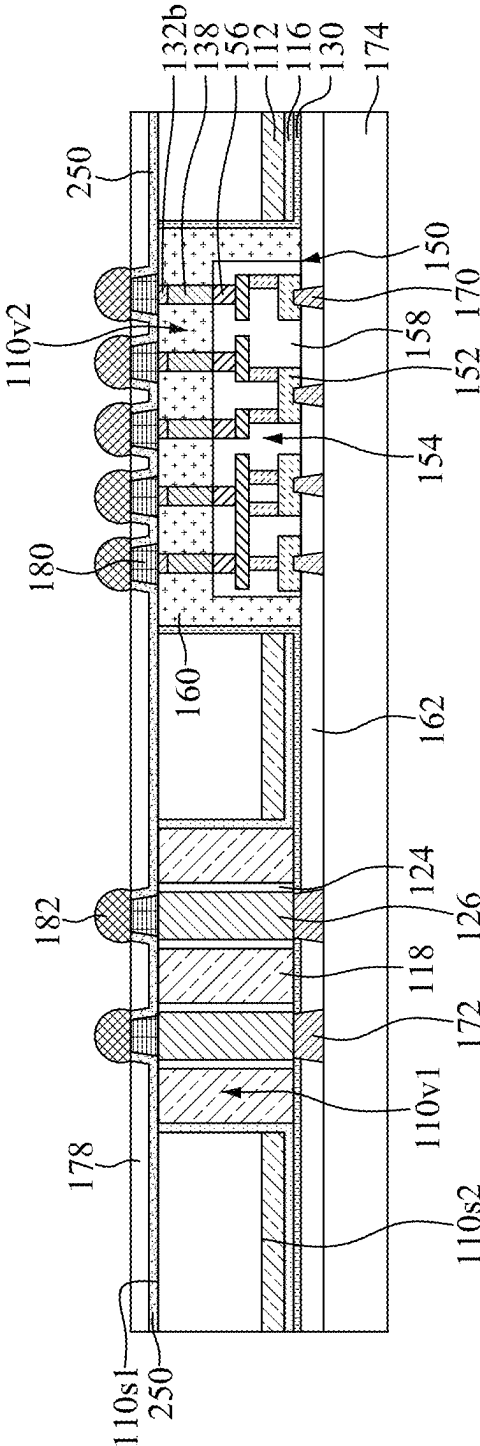


FIG. 29

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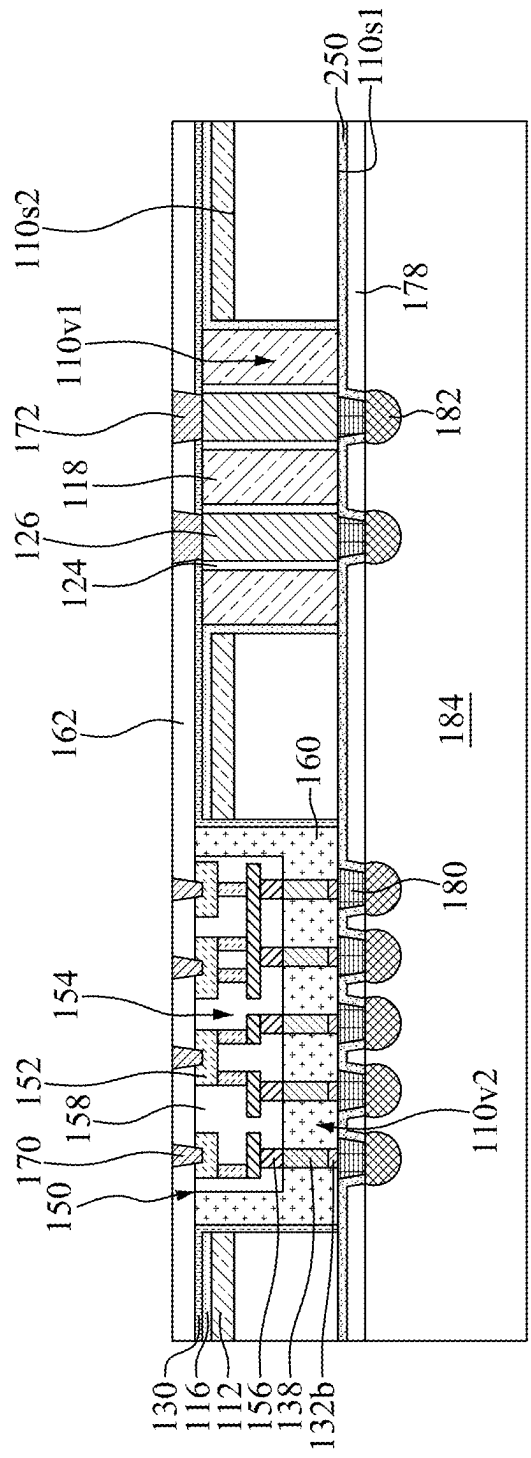


FIG. 30

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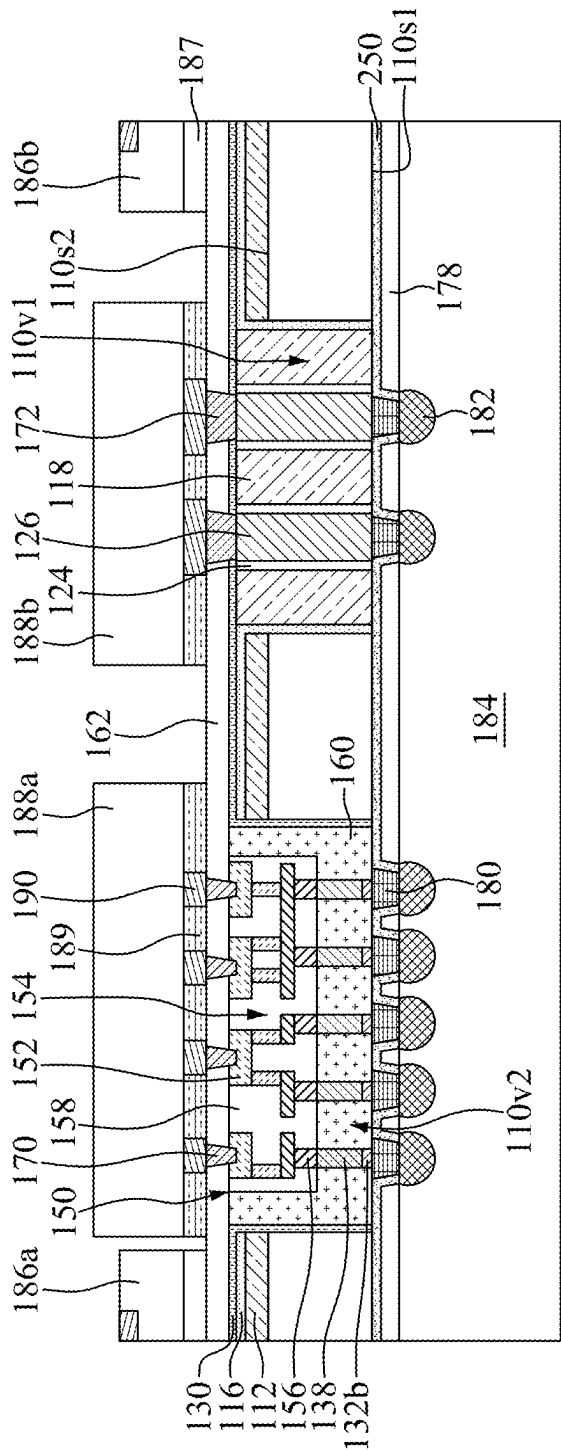


FIG. 31

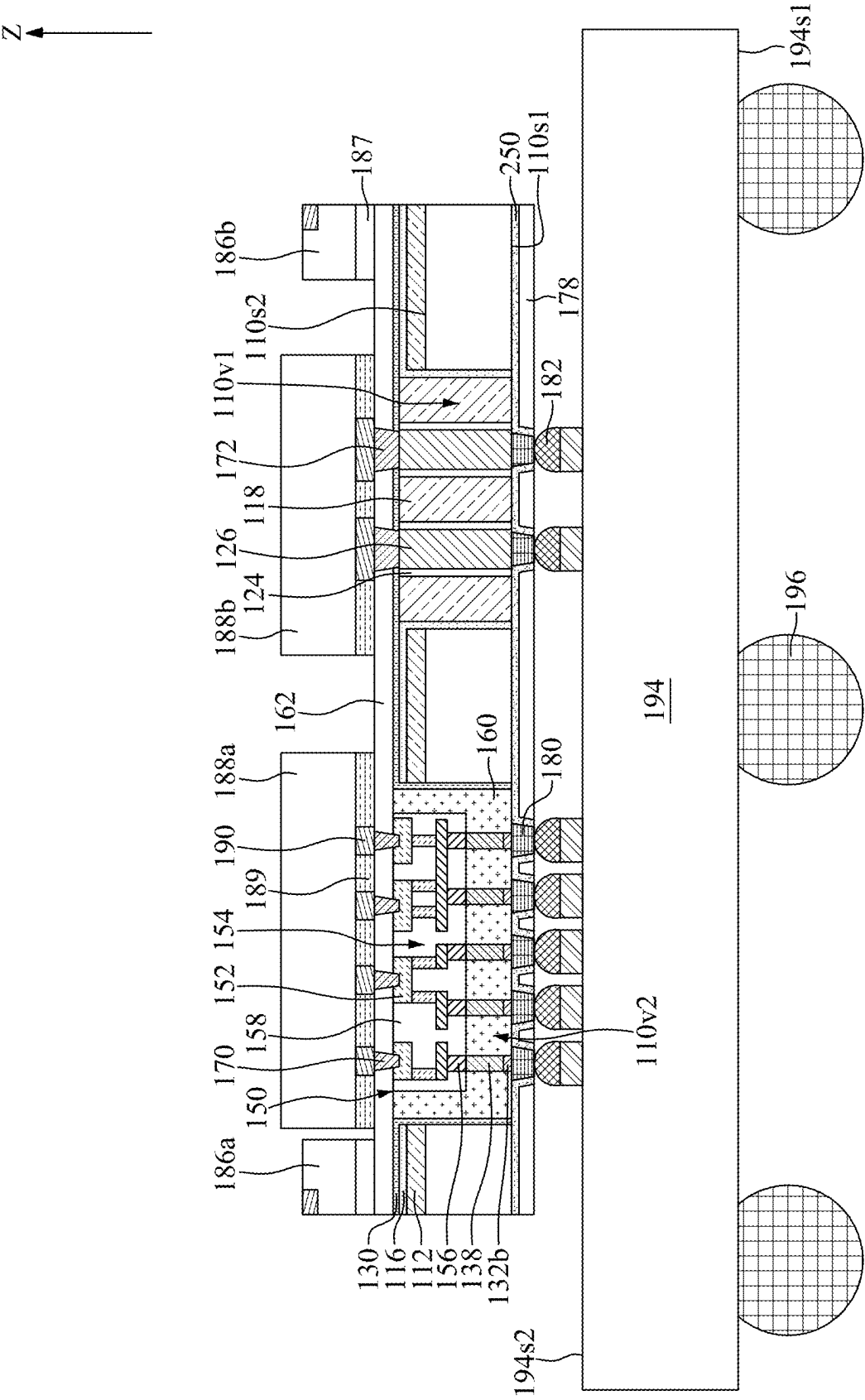


FIG. 32

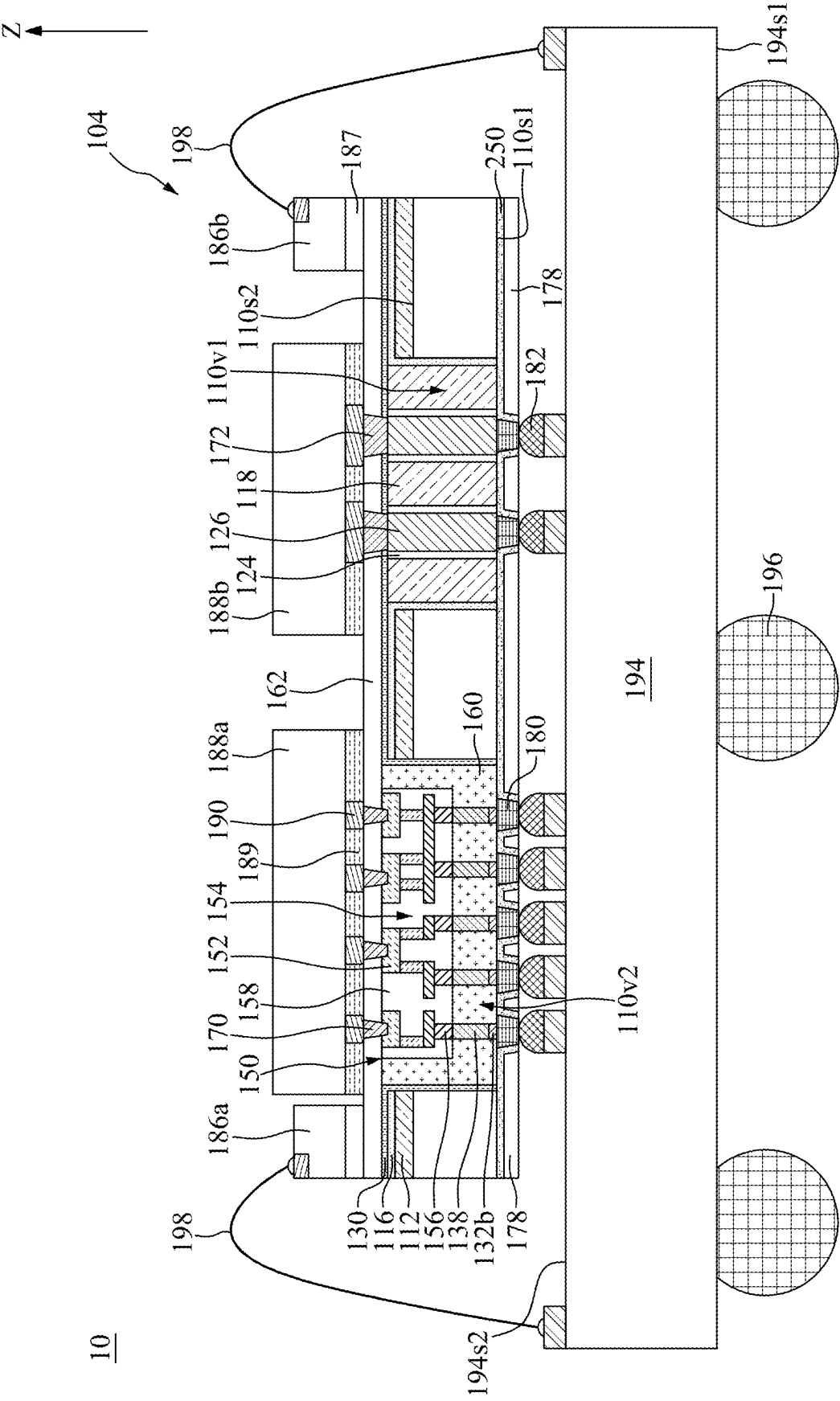


FIG. 33

SEMICONDUCTOR DEVICE WITH SPACER AND METHOD FOR FABRICATING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a divisional application of U.S. Non-Provisional application Ser. No. 18/599,258 filed Mar. 8, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a semiconductor device and a method for fabricating the semiconductor device, and more particularly, to a semiconductor device with a spacer and the method for fabricating the semiconductor device with the spacer.

DISCUSSION OF THE BACKGROUND

[0003] Semiconductor devices are used in a variety of electronic applications, such as personal computers, cellular telephones, digital cameras, and other electronic equipment. The dimensions of semiconductor devices are continuously being scaled down to meet the increasing demand of computing ability. However, a variety of issues arise during the scaling-down process, and such issues are continuously increasing. Therefore, challenges remain in achieving improved quality, yield, performance, and reliability and reduced complexity.

[0004] This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed in this section constitutes prior art to the present disclosure, and no part of this Discussion of the Background section may be used as an admission that any part of this application, including this Discussion of the Background section, constitutes prior art to the present disclosure.

SUMMARY

[0005] One aspect of the present disclosure provides a semiconductor device including an interposer having a first surface and a second surface parallel to the first surface; a conductive via extending between the first surface and the second surface of the interposer; an insulation layer separating the conductive via from the interposer; a first electronic component positioned on the second surface and electrically connected to the conductive via; a first conductive element positioned on the first surface of the interposer and electrically connected to the conductive via; a spacer conformally positioned on the first surface of the interposer and on sidewalls of the first conductive element; and a passivation layer positioned on the spacer. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0006] Another aspect of the present disclosure provides a semiconductor device including an interposer having a first surface and a second surface parallel to the first surface, wherein the interposer defines a first cavity and a second cavity extending between the first surface and the second surface; a through-insulation via positioned within the first cavity; a first electronic component positioned on the second

surface of the interposer and electrically connected to the through-insulation via; a second electronic component positioned within the second cavity; a first conductive element positioned on the first surface of the interposer and electrically connected to the through-insulation via; a spacer conformally positioned on the first surface of the interposer and on sidewalls of the first conductive element; and a passivation layer positioned on the spacer. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0007] Another aspect of the present disclosure provides a method for fabricating a semiconductor device including providing an interposer including a first surface and a second surface parallel to the first surface; recessing a first cavity from the second surface; forming an insulation layer filling the first cavity; forming a conductive via penetrating the insulation layer; thinning the interposer from the first surface to expose the conductive via; forming a first conductive element on the first surface of the interposer and electrically connected to the conductive via; conformally forming a spacer on the first surface of the interposer and sidewalls of the first conductive element; forming a passivation layer on the spacer; and performing a hybrid bonding technique to bond a first electronic component on the second surface of the interposer and electrically connected to the conductive via. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0008] Due to the design of the semiconductor device of the present disclosure, the through-insulation via (e.g., the conductive via) has a relatively small parasitic capacitance in comparison with a through-substrate via. Further the depth of the through-insulation via may be controlled by determining the depth of a cavity of the interposer, and the protection layer may prevent the interposer from being over polished. As a result, the depth of the through-insulation via may be relatively well controlled in comparison with the through-substrate via. In addition, the RC delay may be reduced by employing the passivation layer and spacers having different dielectric constants. As a result, the performance of the semiconductor device may be improved.

[0009] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions

of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0011] FIG. 1 is a top view of a layout of a semiconductor device, in accordance with some embodiments of the present disclosure;

[0012] FIG. 2 is a cross-sectional view along line A-A' of the semiconductor device, in accordance with some embodiments of the present disclosure;

[0013] FIG. 3 and FIG. 4 are partial enlarged views of regions of FIG. 2;

[0014] FIG. 5 is a flowchart illustrating a method of manufacturing a semiconductor device, in accordance with various aspects of the present disclosure;

[0015] FIG. 6 is a top view of a wafer, in accordance with some embodiments of the present disclosure; and

[0016] FIG. 7 to FIG. 33 illustrate one or more various stages of manufacturing the semiconductor device as shown in FIG. 2, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0018] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] It should be understood that when an element or layer is referred to as being “connected to” or “coupled to” another element or layer, it can be directly connected to or coupled to another element or layer, or intervening elements or layers may be present.

[0020] It should be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. Unless indicated otherwise, these terms are only used to distinguish one element from another element. Thus, for example, a first element, a first component or a first section discussed below could be termed a second element, a second component or a second section without departing from the teachings of the present disclosure.

[0021] Unless the context indicates otherwise, terms such as “same,” “equal,” “planar,” or “coplanar,” as used herein when referring to orientation, layout, location, shapes, sizes, amounts, or other measures do not necessarily mean an exactly identical orientation, layout, location, shape, size, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, amounts, or other measures within acceptable variations that may occur, for example, due to manufacturing processes. The term “substantially” may be used herein to reflect this meaning. For example, items described as “substantially the same,” “substantially equal,” or “substantially planar,” may be exactly the same, equal, or planar, or may be the same, equal, or planar within acceptable variations that may occur, for example, due to manufacturing processes.

[0022] In the present disclosure, a semiconductor device generally means a device which can function by utilizing semiconductor characteristics, and an electro-optic device, a light-emitting display device, a semiconductor circuit, and an electronic device are all included in the category of the semiconductor device.

[0023] It should be noted that, in the description of the present disclosure, above (or up) corresponds to the direction of the arrow of the direction Z, and below (or down) corresponds to the opposite direction of the arrow of the direction Z.

[0024] FIG. 1 is a top view of a layout of a semiconductor device 10, in accordance with some embodiments of the present disclosure.

[0025] In some embodiments, the semiconductor device 10 may include an interposer 110. The interposer 110 may include an elementary semiconductor including silicon or germanium in a single crystal form, a polycrystalline form, or an amorphous form; a compound semiconductor material including at least one of silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor material including at least one of SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and GaInAsP; any other suitable materials; or a combination thereof. In some embodiments, the alloy semiconductor substrate may be a SiGe alloy with a gradient Ge feature in which the Si and Ge composition changes from one ratio at one location to another ratio at another location of the gradient SiGe feature. In another embodiment, the SiGe alloy may be formed over a silicon substrate. In some embodiments, the SiGe alloy may be mechanically strained by another material in contact with the SiGe alloy.

[0026] In some embodiments, the semiconductor device 10 may include at least one region 10a, at least one region 10b, and at least one region 10c on or within the interposer 110.

[0027] In some embodiments, the region 10a may be configured to support an electronic component which has a terminal (e.g., a conductive pad) connected to other devices by a conductive wire.

[0028] In some embodiments, the region 10b may be configured to accommodate a package structure or an electronic component(s) disposed within the interposer 110.

[0029] In some embodiments, the region 10c may be configured to form a through-insulation via (TIV), such as a through-oxide via (TOV), which may electrically connect two devices disposed on two opposite sides of the interposer 110.

[0030] FIG. 2 is a cross-sectional view along line A-A' of the semiconductor device 10, in accordance with some embodiments of the present disclosure. FIG. 3 and FIG. 4 are partial enlarged views of the region 10c and the region 10b, respectively.

[0031] The interposer 110 (or substrate) may have a surface 110s1 (or a lower surface) and a surface 110s2 (or an upper surface) opposite to the surface 110s1. In other words, the surface 110s1 and the surface 110s2 may be parallel to each other.

[0032] The semiconductor device 10 may include a dielectric layer 112 and a protection layer 116. In some embodiments, the dielectric layer 112 may be disposed on the surface 110s2 of the interposer 110. In some embodiments, the protection layer 116 may be disposed on the dielectric layer 112. The protection layer 116 may be spaced apart from the interposer 110 by the dielectric layer 112. The dielectric layer 112 and the protection layer 116 may have different materials. Each of the dielectric layer 112 and the protection layer 116 may include silicon oxide (SiO_x), silicon nitride (Si_xN_y), silicon oxynitride (SiON), or a combination thereof. In some embodiments, the dielectric layer 112 may include or be made of silicon oxide. In some embodiments, the protection layer 116 may include or be made of silicon nitride. In some embodiments, the protection layer 116 may function as a polishing stop layer (e.g., chemical mechanical polishing (CMP) stop layer) or an etching stop layer.

[0033] The interposer 110 may define a cavity 110v1. In some embodiments, the cavity 110v1 may extend between the surface 110s1 and the surface 110s2 of the interposer 110. The protection layer 116 may be disposed on a surface 110s3 of the interposer 110 within the cavity 110v1. In some embodiments, the protection layer 116 may be in contact with the surface 110s3 of the interposer 110 within the cavity 110v1.

[0034] As shown in FIG. 3, the semiconductor device 10 may include an insulation layer 118. The insulation layer 118 may fill in the cavity 110v1 of the interposer 110. The insulation layer 118 may be spaced apart from the surface 110s3 of the interposer 110 by the protection layer 116. In some embodiments, the insulation layer 118 may penetrate the interposer 110. The insulation layer 118 may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. The insulation layer 118 and the protection layer 116 may include different materials. In some embodiments, the insulation layer 118 may include or be made of silicon oxide. The protection layer 116 may include a surface 116s1 (or a lower surface). In some embodiments, the surface 116s1 may be substantially coplanar with the surface 110s1 of the interposer 110.

[0035] The semiconductor device 10 may include a seed layer 124. The seed layer 124 may be disposed on a lateral surface of the insulation layer 118. The seed layer 124 may be spaced apart from the protection layer 116 by the insulation layer 118. The seed layer 124 may include conductive materials, such as tantalum nitride (TaN), titanium, titanium nitride (TiN), or other suitable materials. The seed layer 124 may include a surface 124s1 (or a lower surface). In some embodiments, the surface 124s1 may be substantially coplanar with the surface 110s1 of the interposer 110. In some embodiments, the surface 124s1 and the surface 116s1 may be located at the same elevation (or height).

[0036] The semiconductor device 10 may include a conductive via 126. The conductive via 126 may be disposed on a lateral surface of the seed layer 124. The conductive via 126 may be spaced apart from the insulation layer 118 by the seed layer 124. The protection layer 116 may be disposed between the conductive via 126 and the interposer 110. In some embodiments, the conductive via 126 may fully penetrate the interposer 110. The conductive via 126 may include conductive materials, such as copper (Cu), tungsten (W), silver (Ag), gold (Au), ruthenium (Ru), iridium (Ir), nickel (Ni), osmium (Os), rhodium (Rh), aluminum (Al), molybdenum (Mo), cobalt (Co), alloys thereof, combinations thereof or other suitable materials.

[0037] The conductive via 126 may include a surface 126s1 (or a lower surface). In some embodiments, the surface 126s1 may be substantially coplanar with the surface 124s1 of the seed layer 124. In some embodiments, the surface 126s1 and the surface 110s1 may be located at the same elevation (or height). In some embodiments, the conductive via 126 may also be referred to as a through-insulation via (TIV), such as a through-oxide via (TOV).

[0038] Referring back to FIG. 2, the semiconductor device 10 may include a protection layer 130. In some embodiments, the protection layer 130 may be disposed on the surface 110s2 of the interposer 110. In some embodiments, the protection layer 130 may be disposed on the dielectric layer 112. The protection layer 130 may be spaced apart from the interposer 110 by the dielectric layer 112. The protection layer 130 and the dielectric layer 112 may include different materials. The protection layer 130 may include silicon oxide, silicon nitride, silicon oxynitride, or other suitable materials. In some embodiments, the protection layer 130 may include or be made of silicon nitride. In some embodiments, the protection layer 130 may function as a polishing stop layer (e.g., CMP stop layer) or an etching stop layer. The protection layer 130 may include a surface 130s1 (or a lower surface) and a surface 130s2 (or an upper surface). In some embodiments, the surface 130s1 of the protection layer 130 may be substantially coplanar with the surface 110s1 of the interposer 110. In some embodiments, an elevation (or height) of the surface 130s1 of the protection layer 130 may be equal to that of the surface 116s1 of the protection layer 116. In some embodiments, an elevation (or height) of the surface 130s1 of the protection layer 130 may be equal to that of the surface 126s1 of the conductive via 126. In some embodiments, an elevation (or height) of the surface 130s1 of the protection layer 130 may be equal to that of the surface 124s1 of the seed layer 124.

[0039] The semiconductor device 10 may include a cavity 110v2. In some embodiments, the cavity 110v2 may extend between the surface 110s1 and the surface 110s2 of the interposer 110. The protection layer 130 may be disposed on a surface 110s4 of the interposer 110 within the cavity 110v2. In some embodiments, the protection layer 130 may be in contact with the surface 110s4 of the interposer 110 within the cavity 110v2.

[0040] As shown in FIG. 4, the semiconductor device 10 may include electrical connections 132b. In some embodiments, the electrical connection 132b may be disposed on the bottom of the cavity 110v2. The electrical connection 132b may be exposed from the surface 110s1 of the interposer 110. The electrical connection 132b may include solder material, such as alloys of gold and tin solder or alloys of silver and tin solder.

[0041] The semiconductor device **10** may include conductive pillars **138**. In some embodiments, the conductive pillar **138** may be disposed on the electrical connection **132b**. The conductive pillar **138** may be electrically connected to the electrical connection **132b**. In some embodiments, the conductive pillar **138** may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0042] The semiconductor device **10** may include a package structure **150**. In some embodiments, the package structure **150** may be disposed within the cavity **110v2**. The package structure **150** may include a plurality of electronic components **152**, a redistribution structure **154**, a plurality of conductive pads **156**, and an encapsulant **158**.

[0043] Each of the electronic components **152** may include a memory device, such as a dynamic random access memory (DRAM) device, a one-time programming (OTP) memory device, a static random access memory (SRAM) device, or other suitable memory devices. In some embodiments, the electronic component **152** may include a logic device (e.g., system-on-a-chip (SoC), central processing unit (CPU), graphics processing unit (GPU), application processor (AP), microcontroller, etc.), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) device), a front-end device (e.g., analog front-end (AFE) devices) or other devices. The electronic component **152** may have a surface **152s1** (or a lower surface) and a surface **152s2** (or an upper surface) opposite to the surface **152s1**. The surface **152s1** may face the surface **110s1** of the interposer **110**.

[0044] The redistribution structure **154** may include conductive pad(s), trace(s), via(s), layer(s), or other interconnection(s) therein. For example, the redistribution structure **154** may include one or more transmission lines (e.g., communications cables) and one or more grounding lines and/or grounding planes therein.

[0045] The conductive pad **156** may be electrically connected to the redistribution structure **154**. The conductive pad **156** may be electrically connected to the conductive pillar **138**. In some embodiments, the conductive pad **156** may be bonded to the conductive pillar **138**. In some embodiments, the conductive pad **156** may be exposed from the encapsulant **158**. In some embodiments, the conductive pad **156** may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0046] In some embodiments, the encapsulant **158** may encapsulate the electronic components **152**. The encapsulant **158** may encapsulate the conductive pads **156**. In some embodiments, the encapsulant **158** may be made of molding material that may include, for example, a Novolac-based resin, an epoxy-based resin, a silicone-based resin, or other suitable encapsulant. Suitable fillers may also be included, such as powdered SiO₂.

[0047] The semiconductor device **10** may include an encapsulant **160**. In some embodiments, the encapsulant **160** may be disposed within the cavity **110v2** of the interposer **110**. In some embodiments, the encapsulant **160** may encapsulate the package structure **150**. In some embodiments, the encapsulant **160** may encapsulate the electronic components **152**. In some embodiments, the encapsulant **160** may encapsulate the encapsulant **158**. In some embodiments, the

encapsulant **160** may encapsulate the conductive pillars **138**. In some embodiments, the encapsulant **160** may encapsulate the electrical connections **132b**. In some embodiments, the encapsulant **160** may be in contact with the protection layer **130**. In some embodiments, the encapsulant **160** may be spaced apart from the interposer **110** by the protection layer **130**. In some embodiments, the encapsulant **160** may penetrate the interposer **110**. In some embodiments, the encapsulant **160** may fully penetrate the interposer **110**. In some embodiments, the encapsulant **160** may be made of molding material that may include, for example, a Novolac-based resin, an epoxy-based resin, a silicone-based resin, or other suitable encapsulant. Suitable fillers may also be included, such as powdered SiO₂. The encapsulant **160** may have a surface **160s1** (or upper surface). In some embodiments, the surface **160s1** of the encapsulant **160** may be substantially coplanar with the surface **130s2** of the protection layer **130**.

[0048] Referring back to FIG. 2, the semiconductor device **10** may include a dielectric layer **162**. The dielectric layer **162** may be disposed on the surface **110s2** of the interposer **110**. The dielectric layer **162** may be disposed on the protection layer **130**. The dielectric layer **162** may be disposed on the encapsulant **160**. The dielectric layer **162** may include a dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate (FSG), a low-k dielectric material, combinations thereof, and/or other suitable materials. In some embodiments, the material of the dielectric layer **162** may be the same as that of the protection layer **130**.

[0049] The semiconductor device **10** may include conductive elements **170**. In some embodiments, the conductive element **170** may penetrate the dielectric layer **162**. In some embodiments, the conductive element **170** may penetrate a portion of the electronic component **152**. In some embodiments, the conductive element **170** may penetrate the electronic component **152** from the surface **152s2**. In some embodiments, the conductive element **170** may be electrically connected to the electronic component **152**. In some embodiments, the conductive element **170** may be tapered toward the interposer **110**. The conductive element **170** may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0050] The semiconductor device **10** may include conductive elements **172**. In some embodiments, the conductive element **172** may penetrate the dielectric layer **162**. In some embodiments, the conductive element **172** may penetrate the protection layer **130**. In some embodiments, the conductive element **172** may be electrically connected to the conductive via **126**. In some embodiments, the conductive element **172** may be tapered toward the interposer **110**. In some embodiments, the conductive element **170** and conductive element **172** may be located at the same elevation (or height). In some embodiments, the conductive element **170** and conductive element **172** may have different dimensions (e.g., length, width, or area). The conductive element **172** may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0051] The semiconductor device **10** may include a passivation layer **178**. The passivation layer **178** may be disposed on the surface **110s1** of the interposer **110**. The passivation layer **178** may include a dielectric material, such

as polyimide-isoindoloquinazolidione (PIQ), polybenzoxazole (PBO), benzocyclobutene (BCB), or other suitable materials.

[0052] As shown in FIG. 4, the semiconductor device 10 may include conductive elements 180. In some embodiments, the conductive element 180 may penetrate the passivation layer 178. In some embodiments, the conductive element 180 may be disposed on the surface 110s1 of the interposer 110. In some embodiments, the conductive element 180 may be electrically connected to the electrical connection 132b. In some embodiments, the conductive element 180 may be electrically connected to the conductive via 126. Detailedly, the conductive element 180 may have a surface 180s1 (or upper surface) and a surface 180s2 (or lower surface). The width W1 of the surface 180s1 may be greater than or equal to the width W2 of the surface 180s2. The conductive element 180 may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0053] As shown in FIGS. 2 to 4, the semiconductor device 10 may include spacers 250. In some embodiments, the spacer 250 may be conformally disposed on the surface 110s1 of the interposer 110 to separate the interposer 110 and the passivation layer 178. In some embodiments, the spacer 250 may be conformally disposed on the insulation layer 118 to separate the insulation layer 118 and the passivation layer 178. In some embodiments, the spacer 250 may be conformally disposed on the encapsulant 160 to separate the encapsulant 160 and the passivation layer 178. In some embodiments, the spacer 250 may be conformally disposed on sidewalls 180s3 of the conductive elements 180 to laterally separate the conductive elements 180 and the passivation layer 178.

[0054] In some embodiments, the spacer 250 may have a surface 250s1 (or upper surface) and a surface 250s2 (or lower surface). In some embodiments, the surface 250s1 may be substantially coplanar with the surface 116s1 of the protection layer 116. In some embodiments, the surface 250s1 may be substantially coplanar with the surface 110s1 of the interposer 110. In some embodiments, the surface 250s1 may be substantially coplanar with the surface 124s1 of the seed layer 124. In some embodiments, the surface 250s1 may be substantially coplanar with the surface 126s1 of the conductive via 126. In some embodiments, the surface 250s1 may be substantially coplanar with the surface 180s1 of the conductive element 180. In some embodiments, the surface 250s1 may be substantially coplanar with the surface 130s1 of the protection layer 130. In some embodiments, the surface 250s2 may be substantially coplanar with the surface 180s2 of the conductive element 180.

[0055] In some embodiments, the spacers 250 and the passivation layer 178, separating the conductive elements 180 from each other, have different dielectric constants. Detailedly, the passivation layer 178 can have a first dielectric constant, and the spacers 250 can have a second dielectric constant less than the first dielectric constant. Consequently, an effective dielectric constant of the dielectric layer combining the passivation layer 178 with the spacers 250 can be reduced, thereby reducing resistance-capacitance (RC) delay of the semiconductor device 10. In some embodiments, the spacers 250 and the passivation layer 178 may include oxide-based dielectrics. For example, the spacers 250 may include low-k oxide-based dielectrics, such as

carbon-doped silicon oxide or fluorinated oxide, and the passivation layer 178 may include silicon oxide or silicon dioxide.

[0056] The semiconductor device 10 may include electrical connections 182. The electrical connection 182 may be electrically connected to the conductive element 180. The electrical connection 182 may include solder material, such as alloys of gold and tin solder or alloys of silver and tin solder.

[0057] The semiconductor device 10 may include electronic component 186a and electronic component 186b. In some embodiments, the electronic component 186a and the electronic component 186b may be disposed on the region 10a. In some embodiments, the electronic component 186a and the electronic component 186b may be disposed on the surface 110s2 of the interposer 110. In some embodiments, the electronic component 186a and the electronic component 186b may be disposed on the surface 110s2 of the interposer 110. In some embodiments, the electronic component 186a and the electronic component 186b may be free from overlapping the cavity 110v1 and the cavity 110v2. The electronic component 186a may be attached to the surface 110s2 through an adhesive layer 187. In some embodiments, the active surface (not annotated) of the electronic component 186a (or the electronic component 186b) may face away from the interposer 110. Each of the electronic component 186a and the electronic component 186b may include a memory device, such as a dynamic random access memory (DRAM) device, a one-time programming (OTP) memory device, a static random access memory (SRAM) device, or other suitable memory devices. In some embodiments, each of the electronic component 186a and the electronic component 186b may include a logic device (e.g., system-on-a-chip (SoC), central processing unit (CPU), graphics processing unit (GPU), application processor (AP), microcontroller, etc.), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) device), a front-end device (e.g., analog front-end (AFE) devices) or other devices.

[0058] The semiconductor device 10 may include electronic component 188a and the electronic component 188b. In some embodiments, the electronic component 188a and the electronic component 188b may be disposed on the surface 110s2 of the interposer 110. In some embodiments, the electronic component 188a may be disposed on or above the region 10b. In some embodiments, the electronic component 188a may be electrically connected to the electronic component 152. In some embodiments, the electronic component 188a may be disposed directly above the package structure 150. In some embodiments, the electronic component 188b may be disposed on or above the region 10c. In some embodiments, the electronic component 188b may be electrically connected to the conductive via 126. In some embodiments, the electronic component 188b may be disposed directly over the conductive via 126. In some embodiments, the active surface (not annotated) of the electronic component 188a (or the electronic component 188b) may face the interposer 110. Each of the electronic component 188a and the electronic component 188b may include a memory device, such as a dynamic random access memory (DRAM) device, a one-time programming (OTP) memory device, a static random access memory (SRAM) device, or other suitable memory devices. In some embodiments, each of the electronic component 188a and the electronic component 188b may include a logic device (e.g., system-on-a-

chip (SoC), central processing unit (CPU), graphics processing unit (GPU), application processor (AP), microcontroller, etc.), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) device), a front-end device (e.g., analog front-end (AFE) devices) or other devices.

[0059] The semiconductor device **10** may include a passivation layer **189**. The passivation layer **189** may be disposed on the active surface of the electronic component **188a** (or the electronic component **188b**). The passivation layer **189** may include a dielectric material such as PIQ, PBO, BCB, or other suitable materials.

[0060] The semiconductor device **10** may include conductive elements **190**. In some embodiments, the conductive element **190** may be electrically connected to the electronic component **188a** (or the electronic component **188b**). In some embodiments, the conductive element **190** may be electrically connected to the conductive element **170**. The conductive element **190** may penetrate the passivation layer **189**. The conductive element **190** may include or be made of conductive materials, Cu, W, Ag, Au, Ru, Ir, Ni, Os, Rh, Al, Mo, Co, alloys thereof, combinations thereof or other suitable materials.

[0061] The semiconductor device **10** may include a mother board **194**. The mother board **194** may be disposed on the surface **110s1** of the interposer **110**. The mother board **194** may be electrically connected to the electronic component **188a** through the package structure **150**. The mother board **194** may be electrically connected to the electronic component **188b** through the conductive via **126**. The mother board **194** may be electrically connected to the electronic component **186a** (or the electronic component **186b**). The mother board **194** may be formed of, for example, a printed circuit board, such as a paper-based copper foil laminate, a composite copper foil laminate, or a polymer-impregnated glass-fiber-based copper foil laminate. The mother board **194** may include a redistribution layer (RDL) or traces for electrical connection between components. The mother board **194** may have a surface **194s1** (or a lower surface) and a surface **194s2** (or an upper surface) opposite to the surface **194s1**. The surface **194s2** of the mother board **194** may face the interposer **110**.

[0062] The semiconductor device **10** may include electrical connections **196**. The electrical connection **196** may be disposed on the surface **194s1** of the mother board **194**. Each of the electrical connections **196** may be electrically connected to an external device (not shown). The electrical connection **196** may include solder material, such as alloys of gold and tin solder or alloys of silver and tin solder.

[0063] The semiconductor device **10** may include conductive wires **198**. In some embodiments, the conductive wire **198** may be configured to electrically connect the electronic component **186a** (or the electronic component **186b**) to the mother board **194**. The conductive wire **198** may be connected to the surface **194s2** of the mother board **194**.

[0064] In a comparative semiconductor device, a TSV is disposed within an interposer to electrically connect devices on opposite surfaces of the interposer. When the size of the semiconductor device is reduced, it becomes more difficult to control the depth of the TSV. Further, the parasitic capacitance of the TSV may adversely affect the performance of the semiconductor device. In this embodiment, the semiconductor device **10** includes a TIV (e.g., conductive

via **126**). The TIV may penetrate the interposer. The TIV has a relatively small parasitic capacitance in comparison with a TSV. Further the depth of the TIV may be controlled by determining the depth of the cavity (e.g., **110v1**), and the protection layer (e.g., **116**) may prevent the interposer **110** from being over polished. As a result, the depth of the TIV may be relatively well controlled in comparison with the TSV.

[0065] FIG. **5** is a flowchart illustrating a method **20** of manufacturing a semiconductor device, in accordance with various aspects of the present disclosure.

[0066] The method **20** begins with operation **S11** in which an interposer is provided. The interposer has a lower surface and an upper surface.

[0067] The method **20** continues with operation **S13** in which a first cavity may be formed within the interposer. A first protection layer may be formed within the first cavity. A TIV may be formed within the first cavity.

[0068] The method **20** continues with operation **S15** in which a second cavity may be formed within the interposer. A second protection layer may be formed within the second cavity. A package structure, including first electronic components, may be disposed within the second cavity.

[0069] The method **20** continues with operation **S17** in which a polishing technique may be performed to remove a portion of the lower surface of the interposer, the first protection layer, and the second protection layer to expose the TIV.

[0070] The method **20** continues with operation **S19** in which a conductive element may be formed on the lower surface of the interposer and electrically connected to the TIV.

[0071] The method **20** continues with operation **S21** in which spaces are conformally formed on the lower surface of the interposer and sidewalls of the conductive element.

[0072] The method **20** continues with operation **S23** in which a passivation layer may be formed on the spacers. The passivation layer may have different dielectric constant from the spacers.

[0073] The method **20** continues with operation **S25** in which a second electronic component, a third electronic component, and a fourth electronic component are attached to the upper surface of the interposer. The second electronic component may be electrically connected to the TIV. The third electronic component may be electrically connected to the package structure.

[0074] The method **20** continues with operation **S27** in which a mother board is attached to the lower surface of the interposer. A conductive wire is formed to electrically connect the fourth electronic component and the mother board.

[0075] The method **20** is merely an example and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, or after each operation of the method **20**, and some operations described can be replaced, eliminated, or reordered for additional embodiments of the method. In some embodiments, the method **20** can include further operations not depicted in FIG. **5**. In some embodiments, the method **20** can include one or more operations depicted in FIG. **5**.

[0076] FIG. **6** is a top view of a wafer, in accordance with some embodiments of the present disclosure. The wafer **102** may be sawed along scribe lines into a plurality of regions **104**. Each of the regions **104** may correspond to a semicon-

ductor die after the wafer **102** is sawed. The region **104** may include a substrate (or an interposer), and a plurality of active components and/or passive components may be attached to or formed on the substrate. The active component may include a memory die (e.g., dynamic random access memory (DRAM) die, a static random access memory (SRAM) die, etc.), a power management die (e.g., power management integrated circuit (PMIC) die), a logic die (e.g., system-on-a-chip (SoC), central processing unit (CPU), graphics processing unit (GPU), application processor (AP), microcontroller, etc.), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies)) or other active components. The passive component may include a capacitor, a resistor, an inductor, a fuse or other passive components.

[0077] FIG. 7 to FIG. 33 illustrate one or more various stages of manufacturing a semiconductor device **10** as shown in FIG. 2, in accordance with some embodiments of the present disclosure. FIG. 7 to FIG. 31 illustrate cross-sectional views of the region **104** at the stages before the wafer **102** is sawed. FIG. 32 and FIG. 33 illustrate cross-sectional views of the region **104** at the stages after the wafer **102** is sawed.

[0078] Referring to FIG. 7, an interposer **110** may be provided. The interposer **110** may have a surface **110s1** and a surface **110s2**. In some embodiments, a dielectric layer **112** may be formed on the surface **110s2** of the interposer **110**. The dielectric layer **112** may be formed by chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or other suitable techniques. In some embodiments, a mask **114** may be formed on the dielectric layer **112**. The mask **114** may include a positive-tone or negative-tone photoresist such as a polymer. The mask **114** may be patterned to have an opening, which may be configured to define a cavity **110v1** of the interposer **110**. In some embodiments, a photolithography technique and an etching technique may be performed to remove a portion of the interposer **110**. In some embodiments, the cavity **110v1** may be recessed from the surface **110s2** of the interposer **110**. The cavity **110v1** may have a depth **D1** between the surface **110s2** of the interposer **110** and the bottom **110b1** of the cavity **110v1**.

[0079] Referring to FIG. 8, the mask **114** may be removed, and a protection layer **116** may be formed on the dielectric layer **112**. The protection layer **116** may be formed on the surface **110s2** of the interposer **110**. The protection layer **116** may be formed on the surface **110s3** of the interposer **110**. The protection layer **116** may be formed on the bottom **110b1** of the cavity **110v1**. An insulation layer **118** may be formed on the protection layer **116** and fill the cavity **110v1**. Each of the protection layer **116** and insulation layer **118** may be formed by CVD, ALD, PVD, or other suitable techniques. In some embodiments, a polishing technique, such as a chemical mechanical polishing (CMP) may be performed on the insulation layer **118** to expose the protection layer **116**.

[0080] Referring to FIG. 9, a mask **120** may be formed over the protection layer **116** and the insulation layer **118**. The mask **120** may include a positive-tone or negative-tone photoresist such as a polymer. The mask **120** may be patterned to form openings **122**. Each of the openings **122**

may penetrate the insulation layer **118**. The opening **122** may be configured to define the region of a TIV or a TOV.

[0081] Referring to FIG. 10, a seed layer **124** may be formed within the opening **122** as shown in FIG. 9. The seed layer **124** may be in contact with the insulation layer **118**. The conductive via **126** may be formed within the opening **122** as shown in FIG. 9. The conductive via **126** may be formed on the seed layer **124**. Each of the conductive via **126** and the seed layer **124** may be formed by CVD, ALD, PVD, electroplating, or other suitable techniques. In some embodiments, the conductive via **126** and the insulation layer **118** may define a TIV. In some embodiments, the conductive via **126** may be formed of (Cu), tungsten (W), silver (Ag), gold (Au), ruthenium (Ru), iridium (Ir), nickel (Ni), osmium (Os), rhodium (Rh), aluminum (Al), molybdenum (Mo), cobalt (Co), alloys thereof, combinations thereof or other suitable materials. In some embodiments, the conductive via **126** and the insulation layer **118** define a TOV. In some embodiments, the conductive via **126** and the seed layer **124** may be formed of the same material.

[0082] Referring to FIG. 11, a mask **128** may be formed on the protection layer **116**. The mask **128** may cover the conductive via **126**. The mask **128** may include a positive-tone or negative-tone photoresist such as a polymer. The mask **128** may be patterned to have an opening, which may be configured to define a cavity **110v2** of the interposer **110**. In some embodiments, a photolithography technique and an etching technique may be performed to remove a portion of the interposer **110** to form the cavity **110v2**. In some embodiments, the cavity **110v2** may be recessed from the surface **110s2** of the interposer **110**. The cavity **110v2** may have a depth **D2** between the surface **110s2** and the bottom **110b2** of the cavity **110v2**. In some embodiments, the depth **D1** (as shown in FIG. 7) may be substantially equal to the depth **D2**.

[0083] Referring to FIG. 12, the mask **128** may be removed, and a protection layer **130** may be formed on the protection layer **116**. The protection layer **130** may be formed on the surface **110s2** of the interposer **110**. The protection layer **130** may be formed on the surface **110s4** of the interposer **110**. The protection layer **130** may be formed on the bottom **110b2** of the cavity **110v2**. A conductive material **132a** may be formed to fill the cavity **110v2**. The conductive material **132a** may include solder materials, such as alloys of gold and tin solder or alloys of silver and tin solder. The protection layer **130** may be formed by CVD, ALD, PVD, or other suitable techniques. The conductive material **132a** may be formed by, for example, a coating technique, or other suitable techniques. A mask **134** may be formed on the conductive material **132a**. The mask **134** may cover the conductive via **126**. The mask **134** may include a positive-tone or negative-tone photoresist such as a polymer. The mask **134** may be patterned to have openings **136**. In some embodiments, the openings **136** may be located on or within the cavity **110v2**.

[0084] Referring to FIG. 13, conductive pillars **138** may be formed within the openings **136**. In some embodiments, the conductive pillars **138** may be formed within the cavity **110v2**. The conductive pillar **138** may be formed by CVD, ALD, PVD, electroplating, or other suitable techniques.

[0085] Referring to FIG. 14, the mask **134** may be removed. The conductive material **132a** may be exposed.

[0086] Referring to FIG. 15, the conductive material 132a exposed from the conductive pillar 138 may be removed, and the electrical connections 132b may be formed on the protection layer 130.

[0087] Referring to FIG. 16, a package structure 150 may be attached to the conductive pillar 138. In some embodiments, the package structure 150 may be disposed within the cavity 110v2 of the interposer 110. The package structure 150 may include a plurality of electronic components 152, a redistribution structure 154, a plurality of conductive pads 156, and an encapsulant 158 encapsulating the electronic component 152, the redistribution structure 154, and the conductive pads 156. In some embodiments, the conductive pads 156 may be bonded to the conductive pillars 138.

[0088] Referring to FIG. 17, an encapsulant 160 may be formed within the cavity 110v2 to encapsulate the package structure 150.

[0089] Referring to FIG. 18, a dielectric layer 162 may be formed on the package structure 150. The dielectric layer 162 may cover the protection layer 130. The dielectric layer 162 may be formed by CVD, ALD, PVD, or other suitable techniques.

[0090] Referring to FIG. 19, a mask 164 may be formed on the dielectric layer 162. The mask 164 may include a positive-tone or negative-tone photoresist such as a polymer. The mask 164 may be patterned to have openings 166 and openings 168. In some embodiments, the opening 166 may penetrate the mask 164 and the dielectric layer 162. In some embodiments, the opening 166 may penetrate a portion of the electronic component 152. In some embodiments, the opening 166 may be located above the cavity 110v2. In some embodiments, the opening 168 may penetrate the mask 164 and the dielectric layer 162. In some embodiments, the opening 168 may penetrate the protection layer 130. In some embodiments, the opening 168 may be located above the cavity 110v1. The conductive via 126 may be exposed from the opening 168.

[0091] Referring to FIG. 20, the mask 164 may be removed, and conductive elements 170 and conductive elements 172 may be formed. The conductive element 170 may be formed within the opening 166 as shown in FIG. 19. The conductive element 172 may be formed within the opening 168 as shown in FIG. 19. In some embodiments, a conductive layer (not shown) may be formed, for example, by CVD, ALD, PVD, electroplating, or other suitable techniques, on the dielectric layer 162. The conductive layer may fill the opening 166 and the opening 168, and a CMP technique may be performed to remove the excessive portion of the conductive layer. As a result, the conductive elements 170 and the conductive element 172 may be formed.

[0092] Referring to FIG. 21, the interposer 110 may be attached to a carrier 174. The carrier 174 may be in contact with the dielectric layer 162. The carrier 174 may include a glass substrate, a ceramic substrate, a plastic substrate, or other suitable carriers.

[0093] Referring to FIG. 22, a grinding technique, such as a CMP technique may be performed on the surface 110s1 of the interposer 110. In some embodiments, the protection layer 116 may serve as a polishing stop layer (e.g., CMP stop layer). In some embodiments, the protection layer 130 may serve as a polishing stop layer (e.g., CMP stop layer). The protection layers 116 and protection layer 130 may be exposed from the interposer 110.

[0094] Referring to FIG. 23, a sacrificial layer 176 may be formed on the surface 110s1 of the interposer 110. The sacrificial layer 176 may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof. The sacrificial layer 176 may be formed by CVD, ALD, PVD, electroplating, or other suitable techniques.

[0095] Referring to FIG. 24, a polishing technique, such as a CMP technique may be performed to remove the sacrificial layer 176. In some embodiments, the bottom of the protection layer 116 may be removed. In some embodiments, the bottom of the protection layer 130 may be removed. In some embodiments, a portion of the seed layer 124 may be removed, and the bottom of the conductive via 126 may be exposed. The surface 110s1 of the interposer 110 may be exposed.

[0096] Referring to FIG. 25, conductive elements 180 may be formed on the surface 110s1 of the interposer 110. The conductive element 180 may be electrically connected to the electrical connection 132b and the conductive via 126, respectively and correspondingly. In some embodiments, the conductive element 180 may be tapered away from the interposer 110. That is, the sidewalls 180s3 of the conductive elements 180 may be slanted. In some embodiments, the sidewalls 180s3 of the conductive element 180 may be substantially vertical.

[0097] Referring to FIG. 26, a layer of spacer material 251 may be conformally formed on the surface 110s1 of the interposer 110 and may cover the conductive element 180. In some embodiments, the spacer material 251 may be formed of, for example, a low-k oxide-based dielectric, such as carbon-doped silicon oxide or fluorinated oxide. In some embodiments, the layer of spacer material 251 may be formed by, for example, CVD, ALD, or other suitable techniques.

[0098] Referring to FIG. 27, a passivation layer 178 may be formed on the surface 110s1 of the interposer 110. Detailedly, the passivation layer 178 may be formed on the layer of spacer material 251. State differently, the conductive element 180 and the layer of spacer material 251 may be formed within the passivation layer 178. The passivation layer 178 may be formed by coating or other suitable techniques.

[0099] Referring to FIG. 28, a polishing technique, such as a CMP technique may be performed to remove a portion of the layer of spacer material 251. The surfaces 180s2 of the conductive element 180 may be exposed. The remaining spacer material 251 may be referred to as the spacers 250. In some embodiments, the surface 180s2 of the conductive element 180, the surface 110s1 of the interposer 110, and the surface 250s2 of the spacer 250 may be substantially coplanar.

[0100] Referring to FIG. 29, in some embodiments, electrical connections 182 may be formed on the conductive elements 180, respectively and correspondingly.

[0101] Referring to FIG. 30, the surface 110s1 of the interposer 110 may be attached to a holder 184. The holder 184 may include a tape, or other suitable materials.

[0102] Referring to FIG. 31, electronic components 186a, 186b, 188a, and 188b may be attached to the surface 110s2 of the interposer 110. The electronic component 186a and the electronic component 186b may be attached to the dielectric layer 162 by an adhesive layer 187. The passivation layer 189 and the conductive elements 190 may be formed on the active surface of the electronic component

188a and the electronic component **188b**. In some embodiments, a hybrid bonding technique may be performed to bond the electronic component **188a** (or the electronic component **188b**) and the dielectric layer **162** and the conductive elements **170** (or the conductive elements **172**). Detailedly, the hybrid bonding technique may include an oxide-to-oxide bonding and a metal-to-metal bonding. The oxide-to-oxide bonding may originate from the bonding between the passivation layer **189** of the electronic component **188a** (or the electronic component **188b**) and the dielectric layer **162**. The metal-to-metal bonding may originate from the bonding between the conductive elements **190** of the electronic component **188a** (or the electronic component **188b**) and the conductive elements **170** (or the conductive elements **172**). In some embodiments, a temperature of hybrid bonding technique may be between about 300° C. and about 450° C.

[0103] In some embodiments, the electronic component **188a** may be disposed above the package structure **150**. The electronic component **188b** may be disposed above the conductive via **126**.

[0104] Referring to FIG. **32**, the holder **184** may be removed. The interposer **110** may be sawed so that a plurality of structures as shown in FIG. **31** may be separated. A mother board **194** may be attached to the surface **110s1** of the interposer **110**. In some embodiments, the electrical connections **182** may be bonded to the pads (not annotated) of the mother board **194**. In some embodiments, the electrical connections **196** may be bonded to the pads (not annotated) of the mother board **194**.

[0105] Referring to FIG. **33**, conductive wires **198** may be formed to electrically connect the mother board **194** and the electronic component **186a** (or **186b**). As a result, the semiconductor device **10** may be produced.

[0106] One aspect of the present disclosure provides a semiconductor device including an interposer having a first surface and a second surface parallel to the first surface; a conductive via extending between the first surface and the second surface of the interposer; an insulation layer separating the conductive via from the interposer; a first electronic component positioned on the second surface and electrically connected to the conductive via; a first conductive element positioned on the first surface of the interposer and electrically connected to the conductive via; a spacer conformally positioned on the first surface of the interposer and on sidewalls of the first conductive element; and a passivation layer positioned on the spacer. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0107] Another aspect of the present disclosure provides a semiconductor device including an interposer having a first surface and a second surface parallel to the first surface, wherein the interposer defines a first cavity and a second cavity extending between the first surface and the second surface; a through-insulation via positioned within the first cavity; a first electronic component positioned on the second surface of the interposer and electrically connected to the through-insulation via; a second electronic component positioned within the second cavity; a first conductive element positioned on the first surface of the interposer and electrically connected to the through-insulation via; a spacer conformally positioned on the first surface of the interposer and on sidewalls of the first conductive element; and a

passivation layer positioned on the spacer. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0108] Another aspect of the present disclosure provides a method for fabricating a semiconductor device including providing an interposer including a first surface and a second surface parallel to the first surface; recessing a first cavity from the second surface; forming an insulation layer filling the first cavity; forming a conductive via penetrating the insulation layer; thinning the interposer from the first surface to expose the conductive via; forming a first conductive element on the first surface of the interposer and electrically connected to the conductive via; conformally forming a spacer on the first surface of the interposer and sidewalls of the first conductive element; forming a passivation layer on the spacer; and performing a hybrid bonding technique to bond a first electronic component on the second surface of the interposer and electrically connected to the conductive via. The passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

[0109] Due to the design of the semiconductor device of the present disclosure, the TIV (e.g., the conductive via **126**) has a relatively small parasitic capacitance in comparison with a TSV. Further the depth of the TIV may be controlled by determining the depth of the cavity (e.g., **110v1**), and the protection layer (e.g., **116**) may prevent the interposer **110** from being over polished. As a result, the depth of the TIV may be relatively well controlled in comparison with the TSV. In addition, the RC delay may be reduced by employing the passivation layer **178** and spacers **250** having different dielectric constants. As a result, the performance of the semiconductor device **10** may be improved.

[0110] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

[0111] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, and steps.

What is claimed is:

1. A method for fabricating a semiconductor device, comprising:

providing an interposer comprising a first surface and a second surface parallel to the first surface;
recessing a first cavity from the second surface;
forming an insulation layer filling the first cavity;

forming a conductive via penetrating the insulation layer; thinning the interposer from the first surface to expose the conductive via;

forming a first conductive element on the first surface of the interposer and electrically connected to the conductive via;

conformally forming a spacer on the first surface of the interposer and sidewalls of the first conductive element; forming a passivation layer on the spacer; and

performing a hybrid bonding technique to bond a first electronic component on the second surface of the interposer and electrically connected to the conductive via;

wherein the passivation layer has a first dielectric constant, and the spacer has a second dielectric constant less than the first dielectric constant.

2. The method for fabricating the semiconductor device of claim 1, wherein a temperature of the hybrid bonding technique is between about 300° C. and about 450° C.

3. The method for fabricating the semiconductor device of claim 2, further comprising recessing a second cavity from the second surface.

4. The method for fabricating the semiconductor device of claim 3, further comprising forming a conductive pillar within the second cavity.

5. The method for fabricating the semiconductor device of claim 4, further comprising forming a second electronic

component within the second cavity and electrically connected to the conductive pillar.

6. The method for fabricating the semiconductor device of claim 5, further comprising encapsulating the second electronic component and the conductive pillar by a first encapsulant.

7. The method for fabricating the semiconductor device of claim 6, further comprising bonding a third electronic component on the second electronic component and electrically connected to the second electronic component.

8. The method for fabricating the semiconductor device of claim 2, further comprising conformally forming a first protection layer on the second surface of the interposer and in the first cavity, wherein the insulation layer is formed on the first protection layer.

9. The method for fabricating the semiconductor device of claim 8, wherein the insulation layer comprises silicon oxide.

10. The method for fabricating the semiconductor device of claim 9, wherein the first protection layer comprises silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof.

11. The method for fabricating the semiconductor device of claim 10, wherein the spacer comprises low-k oxide-based dielectrics.

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